

GenAI Curriculum Generator

AI-Powered Educational Curriculum Design with IBM Granite 3.3 2B

Project Description

CurrHub – GenAI Curriculum Generator & Study Companion is an AI-powered educational platform designed to simplify curriculum planning and improve the learning experience for both faculty and students. The system helps faculty generate structured, semester-wise curricula using Generative AI while enabling students to access personalized study roadmaps and academic guidance.

The platform uses the Groq API with the LLaMA 3.3 70B model, along with React.js, Firebase Authentication, and Firestore Database, to generate intelligent and industry-relevant curriculum structures. Based on inputs such as domain, duration, semesters, and learning objectives, the system creates complete course plans including subjects, topics, credits, and learning outcomes.

Faculty members can create, manage, and export curricula, while students can explore curricula and receive AI-powered learning recommendations. The platform also includes an AI Study Companion (Sage AI) that assists students by answering questions, explaining concepts, generating study plans, and suggesting projects.

CurrHub combines Generative AI with modern web technologies to automate curriculum design, reduce manual effort, and support effective learning and academic planning.

Scenarios

Scenario 1 – Faculty Curriculum Generation

A faculty member logs into CurrHub and selects the **Curriculum Generator** module. They enter details such as the course domain (e.g., Artificial Intelligence), number of semesters, course duration, and learning objectives. The AI processes the input and generates a complete semester-wise curriculum containing subjects, topics, credits, and learning outcomes. The faculty member can review, edit, and save the generated curriculum in the database. Finally, the curriculum can be exported as a professional PDF document for academic use and approval purposes.

Scenario 2 – Student Personalized Learning Roadmap

A student accesses the platform and selects a skill or career path they want to learn, such as Data Science, Web Development, or Machine Learning. Based on the selected domain, the AI generates a personalized learning roadmap that outlines the sequence of topics, recommended learning stages, and project suggestions. The roadmap helps students understand what to learn, in which order to learn it, and the skills required at each stage, making self-learning more organized and effective.

Scenario 3 – Sage AI Study Companion

A student wants to learn additional topics alongside their academic curriculum. Using Sage AI, the student enters a learning goal such as Docker, Kubernetes, or Machine Learning. Sage analyzes the student's existing curriculum and generates a personalized roadmap showing how the new topics can be learned alongside current semester subjects.

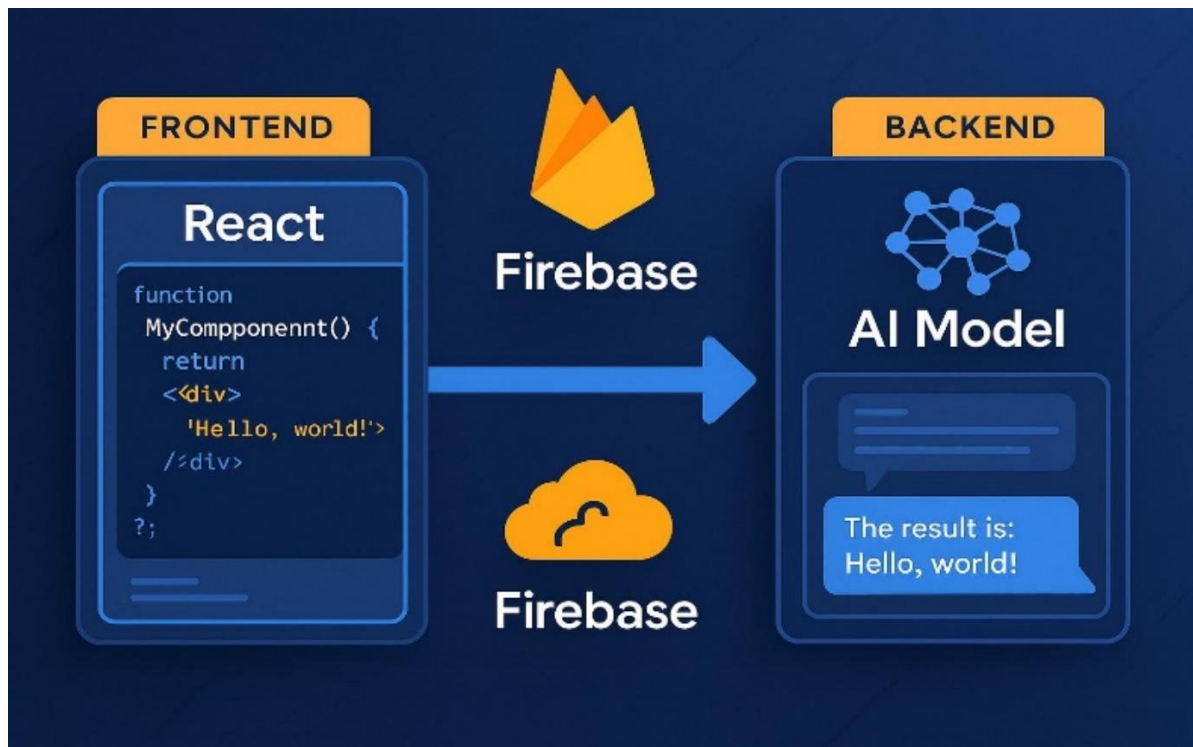
Scenario 4 – Curriculum Management and Sharing

After generating a curriculum, faculty members can store it securely in the Firestore database. They can later retrieve, update, or modify the curriculum whenever changes are required. Students can view the published curricula relevant to their courses and understand the subjects, topics, and learning outcomes planned for each semester. This ensures transparency and keeps both faculty and students aligned with academic objectives.

Scenario 5 – Career-Oriented Curriculum Design

An educational institution wants to design a curriculum aligned with current industry requirements. The faculty selects a domain such as Artificial Intelligence or Full Stack Development and specifies industry-focused objectives. The AI generates a curriculum that includes relevant technologies, practical skills, and project-based learning outcomes. This helps institutions create modern, industry-oriented programs that improve student employability and skill development.

Architecture Overview



CurrHub – GenAI Curriculum Generator & Study Companion is a full-stack web application that combines Artificial Intelligence with cloud-based educational services to simplify curriculum design and academic planning. The architecture is designed for scalability, security, and a smooth user experience for both faculty and students.

The frontend is developed using React.js and Vite, providing a fast and responsive interface. Tailwind CSS is used for modern, mobile-friendly UI design. User authentication is managed through Firebase Authentication, ensuring secure login and role-based access for faculty and students.

Curriculum data, user profiles, and feedback are stored in Firebase Firestore, which provides cloud-based storage and real-time synchronization. The platform integrates with the Groq API and the LLaMA 3.3 70B model to generate intelligent curriculum structures, course descriptions, learning outcomes, and personalized study recommendations.

The Sage AI Study Companion helps students by answering academic questions, creating study plans, suggesting projects, and providing learning guidance. The system also supports PDF export functionality for downloading professional curriculum reports.

Core Technologies

- React.js – Frontend development
- Vite – Fast build and development environment
- Tailwind CSS – Responsive UI design
- Firebase Authentication – Secure login and role management
- Firebase Firestore – Cloud database storage
- Groq API – AI inference and processing
- LLaMA 3.3 70B Model – AI-powered curriculum generation
- React Router – Navigation and routing
- jsPDF – PDF report generation
- Vercel – Deployment and hosting

Key Architectural Features

- AI-powered curriculum generation
- Faculty and student role-based access
- Secure authentication and authorization
- Real-time cloud database management
- Personalized learning roadmap generation
- AI Study Companion for academic assistance
- Curriculum publishing and sharing
- Feedback and curriculum improvement system
- PDF export functionality
- Responsive and mobile-friendly interface

Pre-Requisites

Software Requirements

- Node.js (v18 or higher)
- npm Package Manager
- Visual Studio Code
- Firebase Project Setup
- Groq API Key
- Modern Web Browser (Chrome, Edge, Firefox)

Knowledge Prerequisites

- Basic JavaScript and React.js
- Firebase Authentication and Firestore
- REST API Integration
- HTML, CSS, and Tailwind CSS
- Basic Understanding of Generative AI
- Curriculum Design and Educational Planning Concepts

Hardware Requirements

- Intel i5 / AMD Ryzen 5 Processor or higher
- Minimum 8GB RAM (16GB Recommended)
- 10GB Free Storage Space
- Stable Internet Connection
- Windows, Linux, or macOS Operating System

System Requirements

- Secure User Authentication Setup
- Cloud Database Connectivity
- AI Model Integration through Groq API
- PDF Generation Support
- Responsive Web Environment for Multi-Device Access

Project Workflow

The development of **CurrHub – GenAI Curriculum Generator & Study Companion** follows a structured workflow that integrates Artificial Intelligence, cloud services, and modern web technologies to provide an efficient curriculum management platform. The workflow consists of five major phases, from environment setup to deployment and testing.

Phase 1: Environment Setup & Configuration

This phase focuses on setting up the development environment and configuring the required tools and services.

Step 1: Project Initialization

- Create React project using Vite
- Configure project folder structure
- Install required dependencies
- Set up Tailwind CSS

Step 2: Firebase Configuration

- Create Firebase project
- Enable Firebase Authentication
- Configure Firestore Database
- Connect Firebase services to the application

Step 3: Groq API Integration Setup

- Obtain Groq API key
- Configure environment variables
- Establish connection with LLaMA 3.3 70B model
- Test AI response generation

Phase 2: Core Application Development

This phase involves building the main functionalities of the platform.

Step 1: Authentication Module

- Implement user registration and login
- Create faculty and student roles
- Manage user sessions securely

Step 2: Curriculum Generation Module

- Integrate AI-powered curriculum generation
- Generate semester-wise course structures
- Create learning outcomes and course descriptions

Step 3: Curriculum Management

- Save generated curricula
- Edit and update curriculum details
- Publish curricula for student access

Step 4: PDF Export Functionality

- Generate downloadable curriculum reports
- Format curriculum data professionally
- Enable PDF export for users

Phase 3: Frontend Development

This phase focuses on creating an interactive and responsive user interface.

Step 1: UI Design

- Design faculty and student dashboards
- Create curriculum generation forms
- Develop navigation components

Step 2: Responsive Styling

- Implement Tailwind CSS styling
- Ensure mobile and desktop compatibility
- Create user-friendly layouts

Step 3: Sage AI Study Companion

- Develop AI chat interface
- Provide study recommendations
- Generate personalized learning roadmaps

Phase 4: Integration & Testing

This phase ensures all components work together efficiently.

Step 1: System Integration

- Connect frontend with Firebase services
- Integrate Groq AI functionality
- Validate data flow between modules

Step 2: Functional Testing

- Test authentication features
- Verify curriculum generation accuracy
- Test curriculum publishing and retrieval
- Validate PDF export functionality

Step 3: Performance Testing

- Evaluate AI response time
- Test database operations
- Verify application responsiveness

Phase 5: Deployment & Documentation

This phase focuses on deploying the application and preparing project documentation.

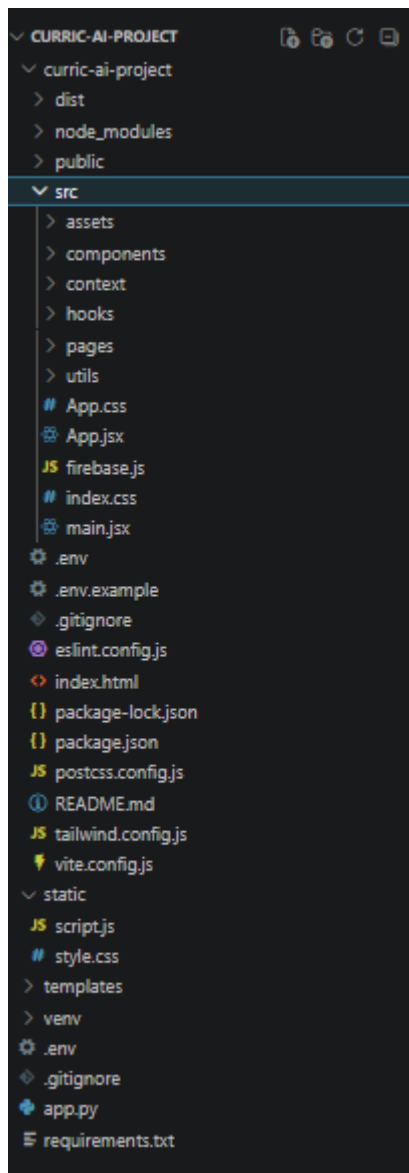
Step 1: Application Deployment

- Build production-ready application
- Deploy on Vercel
- Configure hosting environment

Step 2: Documentation

- Prepare user manual
- Create installation guide
- Document system architecture and workflow
- Develop troubleshooting guide

Project Structure



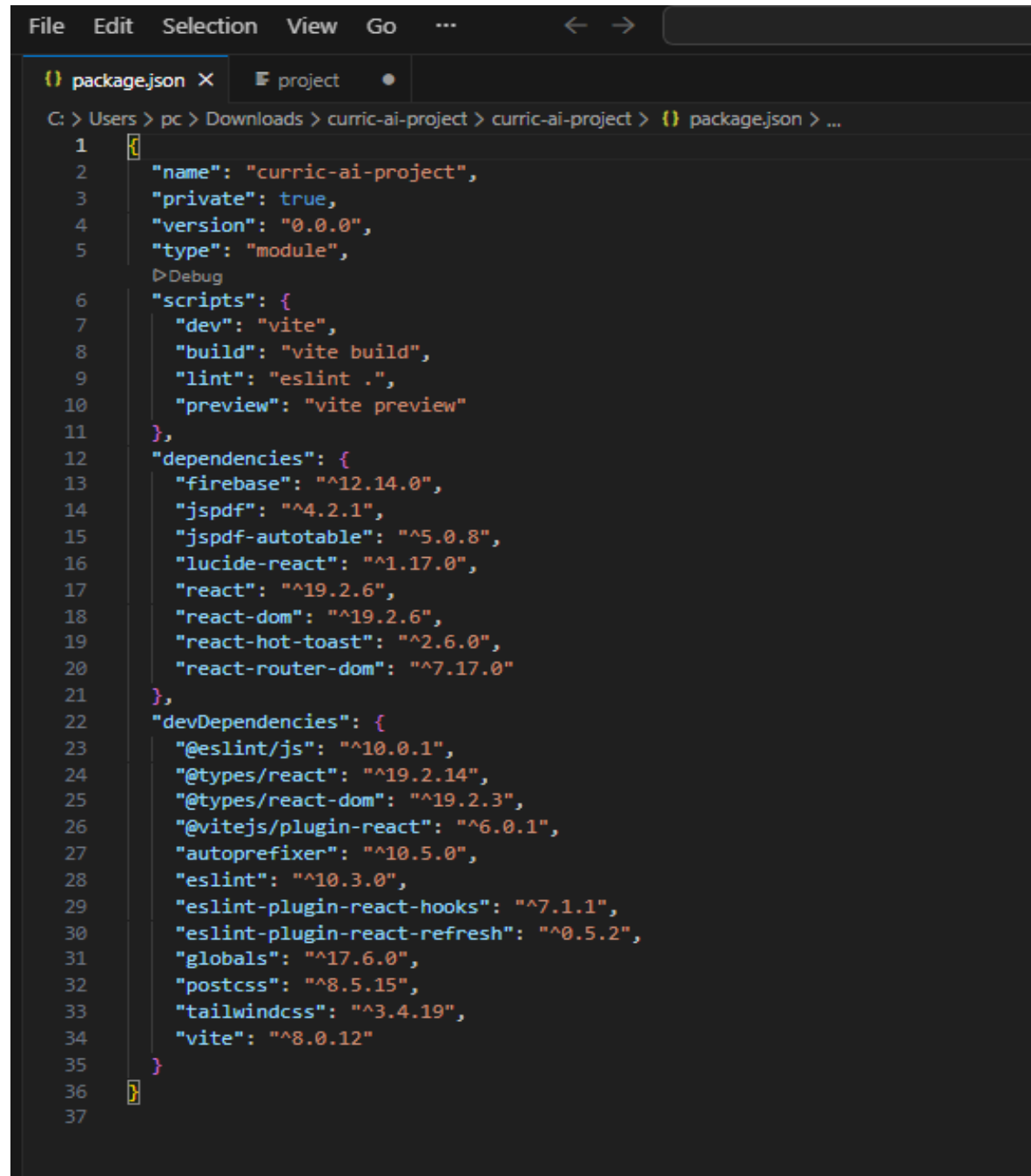
MILESTONE 1: Environment Setup & Configuration

This phase establishes the development environment and configures the essential services required for the CurrHub platform. It includes project initialization, Firebase setup, and AI service integration to ensure a stable foundation for development.

Activity 1.1: Install Dependencies & Setup Environment

The React application is created using Vite, and the required dependencies are installed. The structure is organized to support scalable and maintainable development.

package.json



```
File Edit Selection View Go ...
package.json x project
C:\Users\pc\Downloads\curric-ai-project\curric-ai-project> package.json > ...
1 {
2   "name": "curric-ai-project",
3   "private": true,
4   "version": "0.0.0",
5   "type": "module",
6   "scripts": {
7     "dev": "vite",
8     "build": "vite build",
9     "lint": "eslint .",
10    "preview": "vite preview"
11  },
12  "dependencies": {
13    "firebase": "^12.14.0",
14    "jspdf": "^4.2.1",
15    "jspdf-autotable": "^5.0.8",
16    "lucide-react": "^1.17.0",
17    "react": "^19.2.6",
18    "react-dom": "^19.2.6",
19    "react-hot-toast": "^2.6.0",
20    "react-router-dom": "^7.17.0"
21  },
22  "devDependencies": {
23    "@eslint/js": "^10.0.1",
24    "@types/react": "^19.2.14",
25    "@types/react-dom": "^19.2.3",
26    "@vitejs/plugin-react": "^6.0.1",
27    "autoprefixer": "^10.5.0",
28    "eslint": "^10.3.0",
29    "eslint-plugin-react-hooks": "^7.1.1",
30    "eslint-plugin-react-refresh": "^0.5.2",
31    "globals": "^17.6.0",
32    "postcss": "^8.5.15",
33    "tailwindcss": "^3.4.19",
34    "vite": "^8.0.12"
35  }
36 }
37
```

main.jsx

```
File Edit Selection View Go ...
main.jsx X project
C: > Users > pc > Downloads > curric-ai-project > curric-ai-project > src > main.jsx
1 import { StrictMode } from 'react'
2 import { createRoot } from 'react-dom/client'
3 import './index.css'
4 import App from './App.jsx'
5
6 createRoot(document.getElementById('root')).render(
7   <StrictMode>
8     <App />
9   </StrictMode>,
10 )
11
```

Activity 1.2: Configure Firebase Setup

Firebase services are configured to provide authentication and cloud database support. This enables secure user management and data storage.

src/firebase.js

```
File Edit Selection View Go ...
main.jsx JS firebase.js X project
C: > Users > pc > Downloads > curric-ai-project > curric-ai-project > src > JS firebase.js > ...
1 import { initializeApp } from "firebase/app";
2 import { getAuth } from "firebase/auth";
3 import { getFirestore } from "firebase/firestore";
4
5 const firebaseConfig = {
6   apiKey: import.meta.env.VITE_FIREBASE_API_KEY,
7   authDomain: import.meta.env.VITE_FIREBASE_AUTH_DOMAIN,
8   projectId: import.meta.env.VITE_FIREBASE_PROJECT_ID,
9   storageBucket: import.meta.env.VITE_FIREBASE_STORAGE_BUCKET,
10   messagingSenderId: import.meta.env.VITE_FIREBASE_MESSAGING_SENDER_ID,
11   appId: import.meta.env.VITE_FIREBASE_APP_ID
12 };
13
14 const app = initializeApp(firebaseConfig);
15
16 export const auth = getAuth(app);
17 export const db = getFirestore(app);
18 export default app;
19
```

Activity 1.3: Configure Groq AI Setup

The Groq API and LLaMA model are integrated to enable AI-powered curriculum generation and learning assistance.

src/utls/groqApi.js

```
File Edit Selection View Go ... ← → 🔍 Search ↻

JS groqApi.js • project •
C:\Users\pc> Downloads > curric-ai-project > src > utls > JS groqApi.js > generateCurriculum > userPrompt

1 export async function generateCurriculum({ skill, level, semesters, weeklyHours, industryFocus }) {
2   const apiKey = import.meta.env.VITE_GROQ_API_KEY;
3
4   if (!apiKey || apiKey.trim() === "" || apiKey === "your_groq_api_key_here") {
5     throw new Error("Groq API Key is missing. Please set VITE_GROQ_API_KEY in your .env file.");
6   }
7
8   const systemPrompt = "You are an expert educational curriculum designer. Respond ONLY with valid JSON. No markdown, no explanation.";
9
10  const userPrompt = `
11  Generate a comprehensive, academic-grade curriculum for learning "${skill}" at the "${level}" level, structured across ${semesters} semesters.
12  ${weeklyHours} ? "Design it for approximately ${weeklyHours} hours of weekly workload." : ""
13  ${industryFocus} ? "Align the curriculum with the industry focus area: "${industryFocus}"." : ""
14
15  The output must be a single valid JSON object containing exactly the following structure, with no wrapper markdown, no code block backticks
16  (like \`\`\`json), and no extra comments:
17
18  {
19    "skill": "${skill}",
20    "level": "${level}",
21    "semesters": ${semesters},
22    "weeklyHours": "${weeklyHours || "N/A"}",
23    "industryFocus": "${industryFocus || "General"}",
24    "semestersData": [
25      {
26        "semesterNumber": 1,
27        "courses": [
28          {
29            "courseName": "string",
30            "courseCode": "string (e.g. CS-101)",
31            "credits": number (e.g. 3 or 4),
32            "weeklyHours": number,
33            "description": "string describing the course",
34            "topics": ["array of 4-6 specific topics"],
35            "learningOutcomes": ["array of 3-4 learning outcomes"],
36            "learningResources": [
37              {
38                "title": "string (e.g. name of book, course, or doc)",
39                "url": "string (valid URL or resource link)",
40                "type": "string (e.g. 'Documentation', 'Course', 'Book', 'Video')"
```

```

73     });
74
75     if (!response.ok) {
76       const errData = await response.json().catch(() => ({}));
77       throw new Error(errData?.error?.message || `Groq API returned status ${response.status}`);
78     }
79
80     const data = await response.json();
81     const resultText = data.choices[0].message.content;
82
83     // Parse response
84     const parsedData = JSON.parse(resultText);
85     return parsedData;
86   } catch (error) {
87     console.error("Groq Curriculum Generation Error:", error);
88     throw error;
89   }
90 }
91

```

src/utis/sageApi.js

```

JS sageApi.js • project
C:\Users\pc> Downloads > curric-ai-project > curric-ai-project > src > utis > JS sageApi.js > querySage > contextCurricula.map() callback
1 export async function querySage({ messages, contextCurricula }) {
2   const apiKey = import.meta.env.VITE_GROQ_API_KEY;
3
4   if (!apiKey || apiKey.trim() === "" || apiKey === "your_groq_api_key_here") {
5     throw new Error("Groq API Key is missing. Please set VITE_GROQ_API_KEY in your .env file.");
6   }
7
8   const systemPrompt = `You are Sage, an intelligent academic study companion. Your ONLY purpose is to help students with learning roadmaps,
9 curriculum planning, study strategies, and educational guidance. You have been provided with the student's current curriculum as context.
10 Generate integrated learning roadmaps that show how the new topic fits alongside the existing semester schedule. If asked anything unrelated to education
11
12 You MUST respond ONLY with a valid JSON object matching the following structure:
13 {
14   "type": "message" | "roadmap",
15   "message": "Use this field to respond to conversational queries, general study strategy questions, or when refusing off-topic requests with the
16 exact required refusal string.",
17   "roadmap": {
18     "roadmapTitle": "Title of the integrated learning roadmap",
19     "overview": "Short summary explaining how the new topic/skill fits into the student's existing curriculum context",
20     "weeklyPlan": [
21       {
22         "week": "Week 1",
23         "focus": "Topic focus",
24         "tasks": ["task 1", "task 2"],
25         "resources": ["suggested resource/link 1", "resource 2"]
26       }
27     ],
28     "integrationPoints": [
29       {
30         "semesterNumber": 1,
31         "courseName": "Course Name from current curriculum",
32         "connection": "How this new topic supplements or integrates with this course"
33       }
34     ],
35     "estimatedDuration": "e.g. 8 weeks",
36     "tips": ["study tip 1", "study tip 2"]
37   }
38 }
39

```

If the user is asking a normal question or you are refusing, set "type" to "message" and populate the "message" field. Set "roadmap" to null.
 If the user is asking to learn a new topic, create a roadmap. Set "type" to "roadmap", populate the "roadmap" field, and set "message" to "".

```

};

// Inject current curriculum contexts into the user prompt or system prompt context
let formattedContext = "No curriculum context provided.";
if (contextCurricula && contextCurricula.length > 0) {
  formattedContext = contextCurricula.map((c, i) => {
    return `Curriculum ${i + 1} Context: Skill: ${c.skill}, Level: ${c.level}, Semesters: ${c.semesters}\nJSON Details:\n${JSON.stringify(c.generatedData)}\n`;
  }).join("\n\n");
}

// We append context to the system message or as a separate system-like role message.
const apiMessages = [
  { role: "system", content: systemPrompt },
  { role: "system", content: `Here is the current student curriculum context that you MUST refer to: \n${formattedContext}` },
  ...messages.map(m => ({
    role: m.role,
    content: m.content
  })))
];

try {
  const response = await fetch("https://api.groq.com/openai/v1/chat/completions", {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
      "Authorization": `Bearer ${apiKey}`
    },
    body: JSON.stringify({
      model: "llama-3.3-70b-versatile",
      messages: apiMessages,
      response_format: { type: "json_object" },
      temperature: 0.3
    })
  });

```

```

75   });
76
77   if (!response.ok) {
78     const errData = await response.json().catch(() => ({}));
79     throw new Error(errData?.error?.message || `Groq API returned status ${response.status}`);
80   }
81
82   const data = await response.json();
83   const resultText = data.choices[0].message.content;
84   return JSON.parse(resultText);
85 } catch (error) {
86   console.error("Sage AI Query Error:", error);
87   throw error;
88 }
89 }
90

```

MILESTONE 2: Core Application Development

This phase develops the main functionalities of CurrHub, including user authentication, curriculum generation, curriculum management, and PDF export.

Activity 2.1: Authentication Module

A secure authentication system is implemented for faculty and students using Firebase Authentication. Users can register, log in, and access role-specific features.

User Registration

AuthContext.jsx

```
// Register a new user
async function register(name, email, password, role, college) {
  try {
    const userCredential = await createUserWithEmailAndPassword(auth, email, password);
    const user = userCredential.user;

    // Save user profile in Firestore
    const profileData = {
      uid: user.uid,
      name,
      email,
      role, // 'Faculty' or 'Student'
      college: college.trim(),
      createdAt: new Date().toISOString()
    };

    await setDoc(doc(db, "users", user.uid), profileData);
    setUserProfile(profileData);
    return user;
  } catch (error) {
    console.error("Registration Error:", error);
    throw error;
  }
}
```

User Login

```
// Login existing user
async function login(email, password) {
  try {
    const userCredential = await signInWithEmailAndPassword(auth, email, password);
    // Profile will be loaded by onAuthStateChanged listener
    return userCredential.user;
  } catch (error) {
    console.error("Login Error:", error);
    throw error;
  }
}
```

Custom Authentication Hook:

useAuth.js

```
C:\> Users > pc > Downloads > curric-ai-project > curric-ai-project > src > hooks > JS useA
1  import { useAuthContext } from "../context/AuthContext";
2
3  export function useAuth() {
4    return useAuthContext();
5  }
6
```

Login.jsx

```
27  const handleSubmit = async (e) => {
28    e.preventDefault();
29    const { email, password } = formData;
30
31    if (!email.trim() || !password.trim()) {
32      toast.error("Please fill in all fields.");
33      return;
34    }
35
36    setLoading(true);
37
38    try {
39      const user = await login(email, password);
40
41      // Load user profile from Firestore to determine role
42      const docRef = doc(db, "users", user.uid);
43      const docSnap = await getDoc(docRef);
44
45      if (docSnap.exists()) {
46        const profile = docSnap.data();
47        toast.success(`Welcome back, ${profile.name}!`);
48        if (profile.role === "Faculty") {
49          navigate("/faculty");
50        } else {
51          navigate("/student");
52        }
53      } else {
54        toast.error("User profile document not found. Please contact administration.");
55      }
56    } catch (error) {
57      console.error(error);
58      toast.error(error.message || "Invalid credentials. Please try again.");
59    } finally {
60      setLoading(false);
61    }
62  };
```

Register.jsx

```
const handleSubmit = async (e) => {
  e.preventDefault();
  const { name, email, password, role, college } = formData;

  if (!name.trim() || !email.trim() || !password.trim() || !college.trim()) {
    toast.error("Please fill in all fields.");
    return;
  }

  if (password.length < 6) {
    toast.error("Password must be at least 6 characters long.");
    return;
  }

  setLoading(false);
  setLoading(true);

  try {
    await register(name, email, password, role, college);
    toast.success(`Registered successfully as ${role}!`);
    if (role === "Faculty") {
      navigate("/faculty");
    } else {
      navigate("/student");
    }
  } catch (error) {
    console.error(error);
    toast.error(error.message || "Registration failed. Please try again.");
  } finally {
    setLoading(false);
  }
};
```


Activity 2.2: Curriculum Generation Module

This module allows faculty members to generate AI-powered curricula by providing educational requirements such as domain, semesters, and learning objectives.

Form Submission & Curriculum Generation Function

GenerateCurriculum.jsx

```
const handleGenerate = async (e) => {
  e.preventDefault();
  if (!formData.skill.trim()) {
    toast.error("Please enter a target skill.");
    return;
  }

  setGenerating(true);
  setGeneratedResult(null);

  try {
    const result = await generateCurriculum({
      skill: formData.skill.trim(),
      level: formData.level,
      semesters: parseInt(formData.semesters, 10),
      weeklyHours: formData.weeklyHours ? parseInt(formData.weeklyHours, 10) : null,
      industryFocus: formData.industryFocus.trim() || null
    });

    setGeneratedResult(result);
    toast.success("Curriculum generated successfully!");

    // Save to generationHistory Firestore collection in real-time
    if (userProfile) {
      await addDoc(collection(db, "generationHistory"), {
        facultyUid: userProfile.uid,
        college: userProfile.college,
        inputs: {
          skill: formData.skill.trim(),
          level: formData.level,
          semesters: parseInt(formData.semesters, 10),
          weeklyHours: formData.weeklyHours ? parseInt(formData.weeklyHours, 10) : null,
          industryFocus: formData.industryFocus.trim() || null
        },
        generatedData: result,
        status: "generated",
        createdAt: new Date().toISOString()
      });
    }
  } catch (error) {
    console.error(error);
    toast.error(error.message || "Failed to generate curriculum. Check your API key.");
  } finally {
    setGenerating(false);
  }
}
```

Firestore Save Function

```
const handleSaveToFirestore = async (status) => {
  if (!generatedResult || !userProfile) return;
  setSaving(true);

  try {
    const docData = {
      skill: formData.skill.trim(),
      level: formData.level,
      semesters: parseInt(formData.semesters, 10),
      weeklyHours: formData.weeklyHours ? parseInt(formData.weeklyHours, 10) : null,
      industryFocus: formData.industryFocus.trim() || "General",
      generatedData: generatedResult,
      publishedBy: userProfile.uid,
      publishedByName: userProfile.name,
      college: userProfile.college,
      status, // 'draft' or 'published'
      createdAt: new Date().toISOString(),
      publishedAt: status === "published" ? new Date().toISOString() : null
    };

    await addDoc(collection(db, "curricula"), docData);
    toast.success(status === "published" ? "Curriculum published to college!" : "Saved as draft successfully.");
    navigate("/faculty/my-curricula");
  } catch (error) {
    console.error("Error saving curriculum:", error);
    toast.error("Failed to save curriculum to database.");
  } finally {
    setSaving(false);
  }
};
```

src/utils/groqApi.js

API Key Validation

```
1 export async function generateCurriculum({ skill, level, semesters, weeklyHours, industryFocus }) {
2   const apiKey = import.meta.env.VITE_GROQ_API_KEY;
3
4   if (!apiKey || apiKey.trim() === "" || apiKey === "your_groq_api_key_here") {
5     throw new Error("Groq API Key is missing. Please set VITE_GROQ_API_KEY in your .env file.");
6   }
7 }
```

Curriculum Generation Request

```
try {
  const response = await fetch("https://api.groq.com/openai/v1/chat/completions", {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
      "Authorization": `Bearer ${apiKey}`
    },
    body: JSON.stringify({
      model: "llama-3.3-70b-versatile",
      messages: [
        { role: "system", content: systemPrompt },
        { role: "user", content: userPrompt }
      ],
      response_format: { type: "json_object" },
      temperature: 0.2
    })
  });
}
```

Response Processing

```
const data = await response.json();
const resultText = data.choices[0].message.content;

// Parse response
const parsedData = JSON.parse(resultText);
return parsedData;
} catch (error) {
  console.error("Groq Curriculum Generation Error:", error);
  throw error;
}
```

Activity 2.3: Curriculum Management Module

Generated curricula are stored in Firestore and can be viewed, edited, published, or deleted by faculty members.

src/hooks/useCurricula.js

addDoc() – Save Curriculum

```
// Save draft or publish new curriculum
async function saveCurriculum(curriculumData) {
  if (!userProfile) throw new Error("User profile not found. Cannot save curriculum.");

  const newDoc = {
    ...curriculumData,
    publishedBy: userProfile.uid,
    publishedByName: userProfile.name,
    college: userProfile.college,
    createdAt: new Date().toISOString(),
    publishedAt: curriculumData.status === "published" ? new Date().toISOString() : null
  };

  const docRef = await addDoc(collection(db, "curricula"), newDoc);
  return docRef.id;
}
```

getDocs() -Retrieve Curricula

```
// Load published curricula for the user's college
const q = query(
  collection(db, "curricula"),
  where("college", "==", userProfile.college),
  where("status", "==", "published"),
  orderBy("createdAt", "desc")
);

const unsubscribe = onSnapshot(q, (snapshot) => {
  const data = snapshot.docs.map(doc => ({
    id: doc.id,
    ...doc.data()
  }));
  setCurricula(data);
  setloading(false);
}, (error) => {
  console.error("Error fetching curricula:", error);
  setloading(false);
});
```

updateDoc() – Publish/Update Curriculum

```
// Update curriculum status (draft -> published, etc.)
async function updateStatus(curriculumId, status) {
  const docRef = doc(db, "curricula", curriculumId);
  const updates = {
    status,
    publishedAt: status === "published" ? new Date().toISOString() : null
  };
  await updateDoc(docRef, updates);
}
```

deleteDoc() – Delete Curriculum

```
// Delete curriculum
async function deleteCurriculum(curriculumId) {
  const docRef = doc(db, "curricula", curriculumId);
  await deleteDoc(docRef);
}
```

src/pages/faculty/MyCurricula.jsx

Retrieve Curricula:

```
const q = query(
  collection(db, "curricula"),
  where("publishedBy", "==", userProfile.uid)
);

const unsubscribe = onSnapshot(q, (snapshot) => {
  const data = snapshot.docs.map(doc => ({
    id: doc.id,
    ...doc.data()
  })).sort((a, b) => new Date(b.createdAt || 0) - new Date(a.createdAt || 0));
  setCurricula(data);
  setloading(false);
}, (error) => {
  console.error("Error fetching curricula:", error);
  setloading(false);
});
```

updateDoc() – Publish Curriculum

```
const handlePublish = async (id, e) => {
  e.stopPropagation(); // prevent expanding the row
  try {
    const docRef = doc(db, "curricula", id);
    await updateDoc(docRef, {
      status: "published",
      publishedAt: new Date().toISOString()
    });
    toast.success("Curriculum published successfully!");
  } catch (error) {
    console.error(error);
    toast.error("Failed to publish curriculum.");
  }
};
```

deleteDoc() – Delete Curriculum

```
const handleDelete = async (id, e) => {
  e.stopPropagation(); // prevent expanding
  if (!window.confirm("Are you sure you want to delete this curriculum? This action cannot be undone.")) {
    return;
  }

  try {
    await deleteDoc(doc(db, "curricula", id));
    toast.success("Curriculum deleted.");
    if (expandedRow === id) {
      setExpandedRow(null);
    }
  } catch (error) {
    console.error(error);
    toast.error("Failed to delete curriculum.");
  }
};
```

GenerationHistory.jsx

getDocs()

```
const q = query(
  collection(db, "generationHistory"),
  where("facultyUid", "==", userProfile.uid)
);
const querySnapshot = await getDocs(q);
const list = querySnapshot.docs.map(doc => ({
  id: doc.id,
  ...doc.data()
})).sort((a, b) => new Date(b.createdAt || 0) - new Date(a.createdAt || 0));
setHistory(list);
} catch (error) {
  console.error("Error loading history:", error);
} finally {
  setLoading(false);
}
}
```

Activity 2.4: PDF Export Functionality

This feature enables users to download generated curricula as professional PDF reports.

src/utls/pdfExport.js

PDF Initialization

```
1 import { jsPDF } from "jspdf";
2 import autoTable from "jspdf-autotable";
3
4 export function exportCurriculumToPDF(curriculum) {
5   const { skill, level, semesters, weeklyHours, industryFocus, generatedData, publishedByName, college } = curriculum;
6   const data = generatedData || {};
7   const doc = new jsPDF({
8     orientation: "portrait",
9     unit: "mm",
10    format: "a4"
11  });
12 }
```

Curriculum Data Export Table

```
autoTable(doc, {
  startY: 25,
  head: [
    ["Course Name", "Code", "Credits", "Hours/Wk", "Key Topics"]
  ],
  body: tableBody,
  headStyles: {
    fillColor: [108, 63, 181], // Deep Purple
    textColor: [255, 255, 255],
    fontStyle: "bold"
  },
  styles: {
    fontSize: 9,
    cellPadding: 3,
    overflow: "linebreak"
  },
  columnStyles: {
    0: { cellWidth: 45 }, // Course Name
    1: { cellWidth: 20 }, // Code
    2: { cellWidth: 15 }, // Credits
    3: { cellWidth: 20 }, // Hours
    4: { cellWidth: 80 } // Topics
  },
  margin: { left: 15, right: 15 }
});
```

PDF Save Function

```
// Save the PDF
const filename = `${skill.toLowerCase().replace(/^[a-z0-9]+/g, "-")}-curriculum.pdf`;
doc.save(filename);
}
```

MILESTONE 3: Frontend Development

This phase focuses on creating a responsive and user-friendly interface for faculty and students.

Activity 3.1: Design Responsive User Interface

The landing page, dashboards, and navigation components are developed to provide a smooth user experience.

Landing.jsx

```

project • JS pdfExport.js Landing.jsx X
C:\Users\pc> Downloads > curric-al-project > src > pages > Landing.jsx > Landing
1  import React from "react";
2  import { Link } from "react-router-dom";
3  import { Sparkles, BookOpen, GraduationCap, Compass, Layers, ShieldCheck } from "lucide-react";
4
5  export function Landing() {
6    return (
7      <div className="min-h-screen bg-gradient-to-b from-indigo-50/50 via-white to-indigo-50/20 flex flex-col justify-between">
8        {
9          /* Navbar */
10         <header className="max-w-7xl mx-auto w-full px-6 py-5 flex justify-between items-center">
11           <div className="flex items-center space-x-2">
12             {/* Logo */}
13             <svg className="w-8 h-8 text-indigo-600" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2">
14               <path d="M22 10v6M2 10v6M10 5 10 5-10 5z" />
15               <path d="M6 12v5c0 2 2 3 6 3s6-1 6-3v-5" />
16             </svg>
17             <span className="text-xl font-extrabold tracking-tight text-slate-800">
18               Curr<span className="text-indigo-600">Hub</span>
19             </span>
20           </div>
21           <div className="flex items-center space-x-4">
22             <Link to="/login" className="text-sm font-semibold text-slate-600 hover:text-indigo-600 transition-colors">
23               Sign In
24             </Link>
25             <Link to="/register" className="bg-indigo-600 hover:bg-indigo-700 text-white text-xs font-semibold px-4.5 py-2.5 rounded-xl shadow-md shadow-indi">
26               Get Started
27             </Link>
28           </div>
29         </header>
30
31         <h1 className="text-4xl md:text-6xl font-black text-slate-900 tracking-tight max-w-4xl leading-tight">
32           Transform Skills into <br className="hidden md:inline"/>
33           <span className="bg-clip-text text-transparent bg-gradient-to-r from-indigo-600 to-purple-600">
34             Semester-wise Syllabi
35           </span>
36         </h1>
37
38         <p className="mt-6 text-base md:text-lg text-slate-500 max-w-2xl leading-relaxed">
39           CurrHub leverages generative AI to co-create complete outcome-based curriculum frameworks for faculty and intelligent personalized study companion
40         </p>
41
42         <div className="grid grid-cols-1 md:grid-cols-2 gap-8 mt-16 md:mt-24 max-w-5xl w-full">
43           {/* Faculty Card */}
44           <div className="bg-white border border-slate-100 rounded-3xl p-8 shadow-xs hover:shadow-md transition-all duration-300 text-left flex flex-col ju">
45             <div>
46               <div className="w-12 h-12 rounded-2xl bg-indigo-50 border border-indigo-100 flex items-center justify-center text-indigo-600 mb-6 group-hover">
47                 <GraduationCap size={24} />
48               </div>
49               <h3 className="text-xl font-bold text-slate-800 mb-3">For Faculty Members</h3>
50               <p className="text-sm text-slate-500 leading-relaxed mb-6">
51                 Generate tailored, high-quality course syllabi, credits allocation, weekly workloads, and capstone suggestions based on any target skill se
52               </p>
53               <ul className="space-y-3 mb-8 text-xs text-slate-600 font-medium">
54                 <li className="flex items-center">
55                   <ShieldCheck size={14} className="text-indigo-600 mr-2 shrink-0" />
56                   College-scoped data dashboard
57                 </li>
58                 <li className="flex items-center">
59                   <Layers size={14} className="text-indigo-600 mr-2 shrink-0" />
60                   Save drafts and publish to college
61                 </li>
62                 <li className="flex items-center">
63                   <BookOpen size={14} className="text-indigo-600 mr-2 shrink-0" />
64                   Track comment threads on published syllabi
65                 </li>
66               </ul>
67             </div>
68           </div>
69         </div>
70       </div>
71     );
72   }
73
74   </pre>

```

src/App.jsx

```

project  JS pdfExport.js  Landing.jsx  App.jsx  X
C:\Users\pc\Downloads\curric-ai-project\curric-ai-project\src> App.jsx > ...
92  export function App() {
103    //
110    <Routes>
111      { /* Public Routes */ }
112      <Route path="/" element={ <Landing /> } />
113      <Route path="/login" element={ <Login /> } />
114      <Route path="/register" element={ <Register /> } />
115
116      { /* Protected Faculty Routes */ }
117      <Route path="/faculty" element={ <FacultyLayout /> } >
118        <Route index element={ <FacultyDashboard /> } />
119        <Route path="generate" element={ <GenerateCurriculum /> } />
120        <Route path="my-curricula" element={ <MyCurricula /> } />
121        <Route path="generation-history" element={ <GenerationHistory /> } />
122        <Route path="comments" element={ <FacultyComments /> } />
123      </Route>
124
125      { /* Protected Student Routes */ }
126      <Route path="/student" element={ <StudentLayout /> } >
127        <Route index element={ <StudentDashboard /> } />
128        <Route path="browse" element={ <BrowseCurricula /> } />
129        <Route path="sage" element={ <SageAI /> } />
130        <Route path="sage-history" element={ <SageHistory /> } />
131        <Route path="my-comments" element={ <MyComments /> } />
132        <Route path="view-history" element={ <ViewHistory /> } />
133      </Route>
134
135      { /* Redirect / Fallback */ }
136      <Route path="*" element={ <Navigate to="/" replace /> } />
137    </Routes>
138  </AuthProvider>
139 </BrowserRouter>
140 );
141 }
142
143 export default App;
144

```

Activity 3.2: Develop Styling & Layout

Tailwind CSS is used to create responsive layouts and attractive user interfaces.

tailwind.config.js

```

C: > Users > pc > Downloads > curric-ai-project > curric-ai-project > JS tailwind.config.js > ...
1  /** @type {import('tailwindcss').Config} */
2  export default {
3    content: [
4      './index.html',
5      './src/**/*..{js,ts,jsx,tsx}',
6    ],
7    theme: {
8      extend: {
9        colors: {
10         primary: {
11           DEFAULT: '#6C3FB5',
12           hover: '#5B339C',
13           light: '#F3EEFB',
14         },
15         sidebarBg: '#F8F7FF',
16         published: '#22C55E',
17         draft: '#F59E0B',
18         sage: {
19           DEFAULT: '#1488A6',
20           hover: '#0D9488',
21           light: '#F0FDFA',
22         },
23       },
24       fontFamily: {
25         sans: ['Inter', 'sans-serif'],
26       },
27     },
28   },
29   plugins: [],
30 }
31
  
```

src/index.css

```

C: > Users > pc > Downloads > curric-ai-project > curric-ai-project > src > # index.css > ...
1  @tailwind base;
2  @tailwind components;
3  @tailwind utilities;
4
5  /* Additional custom scrollbar and scroll behaviors if desired */
6  ::-webkit-scrollbar {
7    width: 6px;
8    height: 6px;
9  }
10 ::-webkit-scrollbar-track {
11   background: transparent;
12 }
13 ::-webkit-scrollbar-thumb {
14   background: #cbd5e1;
15   border-radius: 4px;
16 }
17 ::-webkit-scrollbar-thumb:hover {
18   background: #94a3b8;
19 }
20
  
```

Activity 3.3: Build Faculty Dashboard

The Faculty Dashboard serves as the central workspace for faculty members. It provides curriculum generation statistics, published curriculum tracking, student feedback monitoring, recent curriculum history, and student progress tracking. The dashboard enables faculty members to efficiently manage academic curriculum development and monitor student engagement.

Dashboard Statistics

FacultyDashboard.jsx

```

C:\Users\pc\Downloads\curric-ai-project\curric-ai-project\src\pages\faculty> FacultyDashboard.jsx > ...
1  import React, { useState, useEffect } from "react";
2  import { useAuth } from "../../hooks/useAuth";
3  import { collection, query, where, getDocs, limit, orderBy } from "firebase/firestore";
4  import { db } from "../../firebase";
5  import { BookOpen, Sparkles, MessageSquare, History, Award, CheckCircle, FileText, Calendar } from "lucide-react";
6  import { Link, useNavigate } from "react-router-dom";
7
8  export function FacultyDashboard() {
9    const { userProfile } = useAuth();
10    const navigate = useNavigate();
11
12    const [stats, setStats] = useState({
13      totalGenerated: 0,
14      totalPublished: 0,
15      totalComments: 0
16    });
17    const [recentCurricula, setRecentCurricula] = useState([]);
18    const [studentsList, setStudentsList] = useState([]);
19    const [loading, setLoading] = useState(true);
  
```

Statistics Card UI

FacultyDashboard.jsx / StatCard.jsx

Shows important faculty statistics in a visual card format.

```

153  { /* Stats Cards Grid */ }
154  <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
155
156    { /* Total Generated Card */ }
157    <div className="bg-white border border-slate-100 rounded-3xl p-6 shadow-2xs flex items-center space-x-5 hover:shadow-xs transition-shadow duration-
158      <div className="p-4 bg-indigo-50 rounded-2xl text-indigo-600">
159        <History size={24} />
160      </div>
161      <div>
162        <p className="text-xs font-bold text-slate-400 uppercase tracking-wider">Total Generated</p>
163        <h3 className="text-2xl font-black text-slate-800 mt-1">{stats.totalGenerated}</h3>
164        <p className="text-[10px] text-slate-400 mt-0.5">AI attempts recorded</p>
165      </div>
166    </div>
167
  
```

Sidebar Navigation

From FacultySidebar.jsx

Provides quick navigation between faculty modules.

```

194      {/* Main Content Layout */}
195      <div className="grid grid-cols-1 lg:grid-cols-3 gap-8">
196
197        {/* Left 2 Cols: Recent Curricula */}
198        <div className="lg:col-span-2 space-y-6">
199          <div className="flex justify-between items-center">
200            <h3 className="text-lg font-bold text-slate-800 flex items-center">
201              <BookOpen className="mr-2 text-indigo-600" size={20} />
202              Recent Curricula
203            </h3>
204            <Link to="/faculty/my-curricula" className="text-xs font-bold text-indigo-600 hover:underline">
205              View All
206            </Link>
207          </div>
208
209          {recentCurricula.length === 0 ? (
210            <div className="bg-white border border-slate-100 rounded-3xl p-8 text-center shadow-2xs">
211              <FileText size={40} className="mx-auto text-slate-300 mb-3" />
212              <p className="text-sm font-semibold text-slate-500">No curricula created yet</p>
213              <p className="text-xs text-slate-400 mt-1 mb-5">Start by generating your first outline with LLaMA.</p>
214              <Link
215                to="/faculty/generate"
216                className="bg-indigo-600 text-white text-xs font-bold px-4 py-2.5 rounded-xl hover:bg-indigo-700 shadow-sm transition-all"
217              >
218                Launch Generator
219              </Link>
220            </div>
221          ) : (
  
```

Recent Curriculum Display

From FacultyDashboard.jsx

Displays recently created curricula for easy access.

```

222 <div className="space-y-4">
223   {recentCurricula.map((curric) => {
224     <div
225       key={curric.id}
226       onClick={() => navigate(`/faculty/my-curricula`)}
227       className="bg-white border border-slate-100 rounded-2xl p-5 shadow-2xs hover:shadow-xs hover:border-indigo-100 cursor-pointer flex justify
228     >
229       <div className="min-w-0 flex-1 pr-4">
230         <span className="inline-flex text-[10px] font-bold bg-indigo-50 text-indigo-700 px-2 py-0.5 rounded-md mb-2">
231           {curric.level}
232         </span>
233         <h4 className="font-bold text-slate-800 text-sm truncate">{curric.skill}</h4>
234         <div className="flex items-center text-[10px] text-slate-400 mt-1.5 space-x-3">
235           <span>{curric.semesters} Semesters</span>
236           <span>*</span>
237           <span>{curric.weeklyHours || "N/A"} Hours/Wk</span>
238         </div>
239       </div>
240     </div>
241     <div className="flex flex-col items-end shrink-0">
242       <span className="inline-flex px-2 py-0.5 rounded text-[9px] font-bold uppercase"
243         {curric.status === "published"
244           ? "bg-emerald-50 text-emerald-700 border border-emerald-100"
245           : "bg-amber-50 text-amber-700 border border-amber-100"}
246       ></span>
247       {curric.status}
248     </span>
249     <span className="text-[10px] text-slate-400 mt-2 flex items-center">
250       <Calendar size={10} className="mr-1" />
251       {new Date(curric.createdAt).toLocaleDateString()}
252     </span>
253   </div>
254 </div>
255 )}]
  
```

Faculty Navigation Links:

Defines the navigation menu options available to faculty members

```

project  Sidebar.jsx
C: > Users > pc > Downloads > curric-ai-project > curric-ai-project > src > components > Sidebar.jsx > Sidebar
18 export function Sidebar() {
26   const isFaculty = userProfile.role === "Faculty";
27
28   const facultyLinks = [
29     { label: "Dashboard", path: "/faculty", icon: LayoutDashboard },
30     { label: "Generate Curriculum", path: "/faculty/generate", icon: PlusCircle },
31     { label: "My Curricula", path: "/faculty/my-curricula", icon: BookOpen },
32     { label: "Generation History", path: "/faculty/generation-history", icon: History },
33     { label: "Comments", path: "/faculty/comments", icon: MessageSquare }
34   ];
  
```

Sidebar Navigation:

Dynamically generates sidebar navigation links for easy access to faculty modules.

```

122
123     /* Navigation Links */
124     <nav className="flex-1 px-4 space-y-1 overflow-y-auto">
125       {links.map((link) => {
126         const Icon = link.icon;
127         const isActive = location.pathname === link.path;
128         return (
129           <Link
130             key={link.path}
131             to={link.path}
132             onClick={() => setIsOpen(false)}
133             className={`flex items-center px-4 py-3 text-sm font-medium rounded-xl transition-all duration-200 group ${
134               isActive
135               ? "bg-indigo-600 text-white shadow-md shadow-indigo-600/10"
136               : "text-slate-600 hover:bg-white hover:text-indigo-600 hover:shadow-sm"
137             }`}
138           >
139             <Icon
140               size={18}
141               className={`mr-3 shrink-0 transition-transform duration-200 group-hover:scale-105 ${
142                 isActive ? "text-white" : link.color || "text-slate-400 group-hover:text-indigo-600"
143               }`}
144             />
145             <span className="flex-1 truncate">{link.label}</span>
146           </Link>
147         );
148       })}
149     </nav>
150

```

User Profile Information:

Displays logged-in faculty information and associated institution.

```

18  export function Sidebar() {
19    </Link>
20  }
21  </div>
22
23  /* User Info / College Badge */
24  <div className="p-4 mx-4 my-4 bg-white rounded-xl border border-indigo-100/40 shadow-sm">
25    <div className="flex items-center space-x-2.5">
26      <div className="flex items-center justify-center w-10 h-10 rounded-full bg-indigo-50 text-primary font-bold">
27        {userProfile.name?.charAt(0).toUpperCase()}
28      </div>
29      <div className="flex-1 min-w-0">
30        <p className="text-sm font-semibold truncate text-slate-700">{userProfile.name}</p>
31        <p className="text-xs text-slate-400 font-medium capitalize">{userProfile.role}</p>
32      </div>
33    </div>
34    <div className="flex items-center mt-3 pt-2.5 border-t border-slate-100 text-xs text-slate-500 font-medium truncate">
35      <School size={13} className="mr-1.5 text-indigo-500 shrink-0" />
36      <span className="truncate bg-indigo-50/50 px-2 py-0.5 rounded text-indigo-700 border border-indigo-100/30">
37        {userProfile.college}
38      </span>
39    </div>
40  </div>
41

```

Logout Function:

Allows faculty members to securely sign out and return to the login page.

```
const handleLogout = async () => {  
  try {  
    await logout();  
    navigate("/login");  
  } catch (error) {  
    console.error("Logout Error:", error);  
  }  
};
```

Activity 3.4: Build Student Dashboard

The Student Dashboard provides students with access to published curricula, learning progress tracking, Sage AI interactions, and curriculum exploration. It serves as the primary interface for students to browse academic content and monitor their learning journey.

Dashboard Statistics

From StudentDashboard.jsx

Displays important student metrics:

- Available Curricula
- Sage AI Conversations
- Comments Posted

```
project • StudentDashboard.jsx X
C: > Users > pc > Downloads > curric-ai-project > curric-ai-project > src > pages > student > StudentDashboard.jsx > S

1  import React, { useState, useEffect } from "react";
2  import { useAuth } from "../../hooks/useAuth";
3  import { collection, query, where, getDocs, limit } from "firebase/firestore";
4  import { db } from "../../firebase";
5  import { BookOpen, Sparkles, MessageSquare, Compass, Award } from "lucide-react";
6  import { Link, useNavigate } from "react-router-dom";
7
8  export function StudentDashboard() {
9    const { userProfile } = useAuth();
10   const navigate = useNavigate();
11
12   const [stats, setStats] = useState({
13     collegeCurricula: 0,
14     sageConversations: 0,
15     commentsPosted: 0
16   });
17   const [featuredCurricula, setFeaturedCurricula] = useState([]);
18   const [enrolledCurricula, setEnrolledCurricula] = useState([]);
19   const [loading, setLoading] = useState(true);
20
21   useEffect(() => {
22     if (!userProfile) return;
23
24     async function fetchStudentStats() {
25       try {
26         const uid = userProfile.uid;
27         const college = userProfile.college;
28
29         // 1. Curricula available from their college
30         const currQuery = query(
31           collection(db, "curricula"),
32           where("college", "==", college),
33           where("status", "==", "published")
34         );
35         const currSnap = await getDocs(currQuery);
36         const collegeCurriculaCount = currSnap.size;
37
```

```
// 2. Sage conversations started
const sageQuery = query(collection(db, "sageHistory"), where("studentUid", "==", uid));
const sageSnap = await getDocs(sageQuery);
const sageCount = sageSnap.size;

// 3. Comments posted
const commentsQuery = query(collection(db, "comments"), where("studentUid", "==", uid));
const commentsSnap = await getDocs(commentsQuery);
const commentsCount = commentsSnap.size;

setStats({
  collegeCurricula: collegeCurriculaCount,
  sageConversations: sageCount,
  commentsPosted: commentsCount
});
```

Curriculum Browsing Logic

Retrieves published curricula available for the student's college from Firestore Database.

```
// 1. Curricula available from their college
const currQuery = query(
  collection(db, "curricula"),
  where("college", "==", college),
  where("status", "==", "published")
);
const currSnap = await getDocs(currQuery);
const collegeCurriculaCount = currSnap.size;
```

Display Recent Curricula

Displays recently published curricula and allows students to view detailed curriculum information.

```
<div className="grid grid-cols-1 sm:grid-cols-2 gap-6">
  {featuredCurricula.map((curr) => (
    <div
      key={curr.id}
      onClick={() => navigate(`/student/browse?view=${curr.id}`)}
      className="bg-white border border-slate-100 rounded-2xl p-5 shadow-2xs hover:shadow-xs hover:border-teal-100 cursor-pointer flex flex-c
    >
      <div>
        <span className="inline-flex text-[9px] font-bold bg-teal-55 text-teal-700 px-2 py-0.5 rounded-md mb-3">
          {curr.level}
        </span>
        <h4 className="font-bold text-slate-800 text-xs truncate group-hover:text-teal-600 transition-colors">{curr.skill}</h4>
        <p className="text-[10px] text-slate-400 mt-1 truncate">By: {curr.publishedByName || "Faculty"}</p>
      </div>
      <div className="flex justify-between items-center mt-5 pt-3 border-t border-slate-100 text-[10px] text-slate-400">
        <span>{curr.semesters} Semesters</span>
        <span className="text-teal-600 font-bold group-hover:translate-x-1 transition-transform">Study &arr;</span>
      </div>
    </div>
  ))}
</div>
```


Student Sidebar Navigation

From StudentSidebar.jsx

Provides quick navigation to student features such as curriculum browsing and Sage AI.

```

233     /* Featured Curricula */
234     <div className="space-y-6">
235       <div className="flex justify-between items-center">
236         <h3 className="text-lg font-bold text-slate-800 flex items-center">
237           <Compass className="mr-2 text-indigo-605" size={20} />
238           Recent Curricula
239         </h3>
240         <Link to="/student/browse" className="text-xs font-bold text-indigo-600 hover:underline">
241           Browse All
242         </Link>
243       </div>
244
245       {featuredCurricula.length === 0 ? (
246         <div className="bg-white border border-slate-100 rounded-3xl p-8 text-center shadow-2xs">
247           <BookOpen size={40} className="mx-auto text-slate-300 mb-3" />
248           <p className="text-sm font-semibold text-slate-500">No curricula published yet</p>
249           <p className="text-xs text-slate-400 mt-1">Syllabi published by your college faculty will show up here.</p>
250         </div>
251       ) : (
252         <div className="grid grid-cols-1 sm:grid-cols-2 gap-6">
253           {featuredCurricula.map((curr) => (
254             <div
255               key={curr.id}
256               onClick={() => navigate(`/student/browse?view=${curr.id}`)}
257               className="bg-white border border-slate-100 rounded-2xl p-5 shadow-2xs hover:shadow-xs hover:border-teal-100 cursor-pointer flex flex-c
258             >
259               <div>
260                 <span className="inline-flex text-[9px] font-bold bg-teal-55 text-teal-700 px-2 py-0.5 rounded-md mb-3">
261                   {curr.level}
262                 </span>
263                 <h4 className="font-bold text-slate-800 text-xs truncate group-hover:text-teal-600 transition-colors">{curr.skill}</h4>
264                 <p className="text-[10px] text-slate-400 mt-1 truncate">By: {curr.publishedByName || "Faculty"}</p>
265               </div>
266               <div className="flex justify-between items-center mt-5 pt-3 border-t border-slate-100 text-[10px] text-slate-400">
267                 <div className="flex justify-between items-center mt-3 pt-3 border-t border-slate-100 text-[10px] text-slate-400">
268                   <span>{curr.semesters} Semesters</span>
269                   <span className="text-teal-600 font-bold group-hover:translate-x-1 transition-transform">Study &arr;</span>
270                 </div>
271               </div>
272             </div>
273           </div>
274         </div>
275       </div>
276
277     /* Right side area: Quick Helper Box */
278     <div className="space-y-6">
279       <h3 className="text-lg font-bold text-slate-800">Meet Sage AI </h3>
280       <div className="bg-gradient-to-br from-teal-50 to-emerald-50/20 border border-teal-100/60 rounded-3xl p-6 shadow-2xs space-y-4">
281         <p className="text-xs text-slate-600 leading-relaxed">
282           Sage is your academic companion. If you want to learn extra topics like **Docker, Kubernetes, or AWS** alongside your core college curriculum
283         </p>
284         <ol className="list-decimal pl-4 space-y-2 text-xs text-slate-500 font-medium">
285           <li>Navigate to **Sage AI** sidebar.</li>
286           <li>Select your current published curriculum context cards.</li>
287           <li>Type your learning goal in natural language.</li>
288           <li>Get a personalized vertical timeline mapping extra topics to semesters!</li>
289         </ol>
290         <Link
291           to="/student/sage"
292           className="inline-flex items-center text-xs font-bold text-teal-700 hover:text-teal-850 hover:underline pt-2"
293         >
294           Start chatting with Sage &arr;
295         </Link>
296       </div>
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```

BrowseCurricula.jsx

Retrieves published curricula from Firestore and displays them to students.

```
useEffect(() => {
  if (!userProfile?.college) return;

  // Fetch published curricula for this college
  const q = query(
    collection(db, "curricula"),
    where("college", "==", userProfile.college),
    where("status", "==", "published")
  );

  const unsubscribe = onSnapshot(q, (snapshot) => {
    const list = snapshot.docs.map(doc => ({
      id: doc.id,
      ...doc.data()
    })).sort((a, b) => new Date(b.createdAt || 0) - new Date(a.createdAt || 0));
    setCurricula(list);
    setloading(false);
  }, (error) => {
    console.error("Error fetching published curricula:", error);
    setloading(false);
  });

  return unsubscribe;
}, [userProfile?.college]);
```

Search and Filter

Allows students to search curricula and apply filters based on education level.

```
// Client side filtering
const filteredCurricula = curricula.filter(c => {
  const skill = c.skill || "";
  const level = c.level || "";
  const semesters = String(c.semesters);

  const matchesSearch = skill.toLowerCase().includes(searchTerm.toLowerCase());
  const matchesLevel = levelFilter === "" || level === levelFilter;
  const matchesSemesters = semestersFilter === "" || semesters === semestersFilter;

  return matchesSearch && matchesLevel && matchesSemesters;
});
```

Curriculum Card Display

Displays available curricula in a card-based layout.

```

    {filteredCurricula.length === 0 ? (
      <div className="bg-white border border-slate-100 rounded-3xl p-10 text-center shadow-2xs">
        <AlertCircle size={40} className="mx-auto text-slate-300 mb-3 animate-pulse" />
        <p className="text-sm font-semibold text-slate-500">No published curricula found</p>
        <p className="text-xs text-slate-400 mt-1">Try resetting search keywords or filters.</p>
      </div>
    ) : (
      <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
        {filteredCurricula.map((curr) => (
          <CurriculumCard
            key={curr.id}
            curriculum={curr}
            onClick={() => handleViewCurriculum(curr)}
          />
        ))}
      </div>
    )}
  </div>
)}
</div>
);
}
export default BrowseCurricula;

```

View Curriculum Details

Opens the detailed curriculum view when a student selects a curriculum.

```

const handleViewCurriculum = async (curriculum) => {
  setSelectedCurriculum(curriculum);
  setSearchParams({ view: curriculum.id });

  // Save to viewHistory Firestore collection (preventing duplicates)
  if (userProfile) {
    try {
      const q = query(
        collection(db, "viewHistory"),
        where("studentUid", "==", userProfile.uid),
        where("curriculumId", "==", curriculum.id)
      );
      const snap = await getDocs(q);
      if (!snap.empty) {
        const existingDoc = snap.docs[0];
        await updateDoc(doc(db, "viewHistory", existingDoc.id), {
          viewedAt: new Date().toISOString()
        });
      } else {
        await addDoc(collection(db, "viewHistory"), {
          studentUid: userProfile.uid,
          curriculumId: curriculum.id,
          curriculumTitle: curriculum.skill,
          facultyName: curriculum.publishedByName || "Faculty Member",
          college: userProfile.college,
          viewedAt: new Date().toISOString()
        });
      }
    } catch (err) {
      console.error("Error writing to viewHistory:", err);
    }
  }
};

```

StudentSidebar.jsx

Student Navigation Links

Defines the navigation modules available to students.

```
const studentLinks = [
  { label: "Dashboard", path: "/student", icon: LayoutDashboard },
  { label: "Browse Curricula", path: "/student/browse", icon: BookOpen },
  { label: "Sage AI", path: "/student/sage", icon: Sparkles, color: "text-teal-600" },
  { label: "Sage History", path: "/student/sage-history", icon: History },
  { label: "My Comments", path: "/student/my-comments", icon: MessageSquare },
  { label: "View History", path: "/student/view-history", icon: Eye }
];
```

Sidebar Navigation Rendering

Generates the student sidebar menu dynamically.

```
/* Navigation Links */
<nav className="flex-1 px-4 space-y-1 overflow-y-auto">
  {links.map((link) => {
    const Icon = link.icon;
    const isActive = location.pathname === link.path;
    return (
      <Link
        key={link.path}
        to={link.path}
        onClick={() => setIsOpen(false)}
        className={`flex items-center px-4 py-3 text-sm font-medium rounded-xl transition-all duration-200 group ${
          isActive
            ? "bg-indigo-600 text-white shadow-md shadow-indigo-600/10"
            : "text-slate-600 hover:bg-white hover:text-indigo-600 hover:shadow-sm"
        }`}
      >
        <Icon
          size={18}
          className={`mr-3 shrink-0 transition-transform duration-200 group-hover:scale-105 ${
            isActive ? "text-white" : link.color || "text-slate-400 group-hover:text-indigo-600"
          }`}
        />
        <span className="flex-1 truncate">{link.label}</span>
      </Link>
    );
  })}
</nav>
```

Logout Function

Allows students to securely sign out of the system.

```
const handleLogout = async () => {
  try {
    await logout();
    navigate("/login");
  } catch (error) {
    console.error("Logout Error:", error);
  }
};
```

MILESTONE 4: Integration & Testing

This phase integrates all modules and verifies the system's functionality and performance.

Activity 4.1: End-to-End Integration

All frontend, backend, database, and AI modules are connected and tested together.

src/App.jsx

Defines application routing and navigation between public, faculty, and student modules.

```
    </Routes>
    </AuthProvider>
  </BrowserRouter>
);
}
```

```
    <Routes>
      <{/* Public Routes */>
        <Route path="/" element={<Landing />} />
        <Route path="/login" element={<Login />} />
        <Route path="/register" element={<Register />} />

        <{/* Protected Faculty Routes */>
        <Route path="/faculty" element={<FacultyLayout />>
          <Route index element={<FacultyDashboard />} />
          <Route path="generate" element={<GenerateCurriculum />} />
          <Route path="my-curricula" element={<MyCurricula />} />
          <Route path="generation-history" element={<GenerationHistory />} />
          <Route path="comments" element={<FacultyComments />} />
        </Route>

        <{/* Protected Student Routes */>
        <Route path="/student" element={<StudentLayout />>
          <Route index element={<StudentDashboard />} />
          <Route path="browse" element={<BrowseCurricula />} />
          <Route path="sage" element={<SageAI />} />
          <Route path="sage-history" element={<SageHistory />} />
          <Route path="my-comments" element={<MyComments />} />
          <Route path="view-history" element={<ViewHistory />} />
        </Route>

        <{/* Redirect / Fallback */>
        <Route path="*" element={<Navigate to="/" replace />} />
      </Routes>
    </AuthProvider>
  </BrowserRouter>
);
}
```

Faculty Route Protection and Student Route Protection

Ensures only authenticated faculty members can access faculty modules.

Ensures only authenticated students can access student modules.

```
function StudentLayout() {
  const { currentUser, userProfile, loading } = useAuthContext();

  if (loading) {
    return (
      <div className="min-h-screen flex items-center justify-center bg-slate-50">
        <div className="animate-spin rounded-full h-8 w-8 border-b-2 border-teal-600"></div>
      </div>
    );
  }

  if (!currentUser || !userProfile) {
    return <Navigate to="/login" replace />;
  }

  if (userProfile.role !== "Student") {
    return <Navigate to="/faculty" replace />;
  }

  return (
    <div className="min-h-screen bg-slate-50 flex">
      <Sidebar />
      <main className="flex-1 p-6 md:p-8 ml-0 md:ml-64 transition-all duration-300">
        { /* Mobile spacing adjustment for sidebar toggle */ }
        <div className="h-12 md:hidden"></div>
        <Outlet />
      </main>
    </div>
  );
}
```

Faculty & Student Module Integration

Integrates faculty and student functionalities into a unified application workflow.

```
/* Protected Faculty Routes */
<Route path="/faculty" element=<FacultyLayout />>
  <Route index element=<FacultyDashboard /> />
  <Route path="generate" element=<GenerateCurriculum /> />
  <Route path="my-curricula" element=<MyCurricula /> />
  <Route path="generation-history" element=<GenerationHistory /> />
  <Route path="comments" element=<FacultyComments /> />
</Route>

/* Protected Student Routes */
<Route path="/student" element=<StudentLayout />>
  <Route index element=<StudentDashboard /> />
  <Route path="browse" element=<BrowseCurricula /> />
  <Route path="sage" element=<SageAI /> />
  <Route path="sage-history" element=<SageHistory /> />
  <Route path="my-comments" element=<MyComments /> />
  <Route path="view-history" element=<ViewHistory /> />
</Route>
```

Firebase Integration

This code initializes Firebase services used throughout the application. Authentication is used for user login and registration, while Firestore is used to store curricula, comments, user profiles, generation history, and student activities.

src/firebase.js

```
1 import { initializeApp } from "firebase/app";
2 import { getAuth } from "firebase/auth";
3 import { getFirestore } from "firebase/firestore";
4
5 const firebaseConfig = {
6   apiKey: import.meta.env.VITE_FIREBASE_API_KEY,
7   authDomain: import.meta.env.VITE_FIREBASE_AUTH_DOMAIN,
8   projectId: import.meta.env.VITE_FIREBASE_PROJECT_ID,
9   storageBucket: import.meta.env.VITE_FIREBASE_STORAGE_BUCKET,
10  messagingSenderId: import.meta.env.VITE_FIREBASE_MESSAGING_SENDER_ID,
11  appId: import.meta.env.VITE_FIREBASE_APP_ID
12 };
13
14 const app = initializeApp(firebaseConfig);
15
16 export const auth = getAuth(app);
17 export const db = getFirestore(app);
18 export default app;
19
```

AI API Request

Sends curriculum generation requests to the Groq AI service using the selected model and user inputs

```
const response = await fetch("https://api.groq.com/openai/v1/chat/completions", {
  method: "POST",
  headers: {
    "Content-Type": "application/json",
    "Authorization": `Bearer ${apiKey}`
  },
  body: JSON.stringify({
    model: "llama-3.3-70b-versatile",
    messages: [
      { role: "system", content: systemPrompt },
      { role: "user", content: userPrompt }
    ],
    response_format: { type: "json_object" },
    temperature: 0.2
  })
});
```

AI Response Processing

Processes the AI-generated response and returns curriculum.

```
const data = await response.json();
const resultText = data.choices[0].message.content;

// Parse response
const parsedData = JSON.parse(resultText);
return parsedData;
} catch (error) {
  console.error("Groq Curriculum Generation Error:", error);
  throw error;
}
```


Curriculum Generation Function

Encapsulates the curriculum generation logic into a reusable function that can be called from the Faculty Curriculum Generation module.

```

C:\Users > pc > Downloads > curric-al-project > curric-al-project > src > utils > * groqApi.js > generateCurriculum > response
1  export async function generateCurriculum({ skill, level, semesters, weeklyHours, industryFocus }) {
2      const apiKey = import.meta.env.VITE_GROQ_API_KEY;
3
4      if (!apiKey || apiKey.trim() === "" || apiKey === "your_groq_api_key_here") {
5          throw new Error("Groq API Key is missing. Please set VITE_GROQ_API_KEY in your .env file.");
6      }
7
8      const systemPrompt = "You are an expert educational curriculum designer. Respond ONLY with valid JSON. No markdown, no explanation.";
9
10     const userPrompt = `
11     Generate a comprehensive, academic-grade curriculum for learning "${skill}" at the "${level}" level, structured across ${semesters} semesters.
12     ${weeklyHours ? `Design it for approximately ${weeklyHours} hours of weekly workload.` : ""}
13     ${industryFocus ? `Align the curriculum with the industry focus area: "${industryFocus}"` : ""}
14
15     The output must be a single valid JSON object containing exactly the following structure, with no wrapper markdown, no code block backticks (like ````)
16
17     {
18       "skill": "${skill}",
19       "level": "${level}",
20       "semesters": ${semesters},
21       "weeklyHours": "${weeklyHours || "N/A"}",
22       "industryFocus": "${industryFocus || "General"}",
23       "semestersData": [
24         {
25           "semesterNumber": 1,
26           "courses": [
27             {
28               "courseName": "string",
29               "courseCode": "string (e.g. CS-101)",
30               "credits": number (e.g. 3 or 4),
31               "weeklyHours": number,
32               "description": "string describing the course",
33               "topics": ["array of 4-6 specific topics"],
34               "learningOutcomes": ["array of 3-4 learning outcomes"],
35               "learningResources": [
36                 {
37                   "title": "string (e.g. name of book, course, or doc)",
38                   "url": "string (valid URL or resource link)",
39                   "type": "string (e.g. 'Documentation', 'Course', 'Book', 'Video')",
40                 }
41               ]
42             }
43           ]
44         }
45       ],
46       "capstoneProject": {
47         "title": "string",
48         "description": "string",
49         "deliverables": ["array of 3-4 deliverables"]
50       }
51     }
52
53     Ensure there are exactly ${semesters} semester items in the semestersData array, and each semester has 2-4 comprehensive courses. Ensure each course has 2-4
54     `;
55
56     try {
57       const response = await fetch("https://api.groq.com/openai/v1/chat/completions", {
58         method: "POST",
59         headers: {
60           "Content-Type": "application/json",
61           "Authorization": `Bearer ${apiKey}`
62         },
63         body: JSON.stringify({
64           model: "llama-3.3-70b-versatile",
65           messages: [
66             { role: "system", content: systemPrompt },
67             { role: "user", content: userPrompt }
68           ],
69           response_format: { type: "json_object" },
70           temperature: 0.2
71         })
72       });
73     }
  
```

```
    })
  });

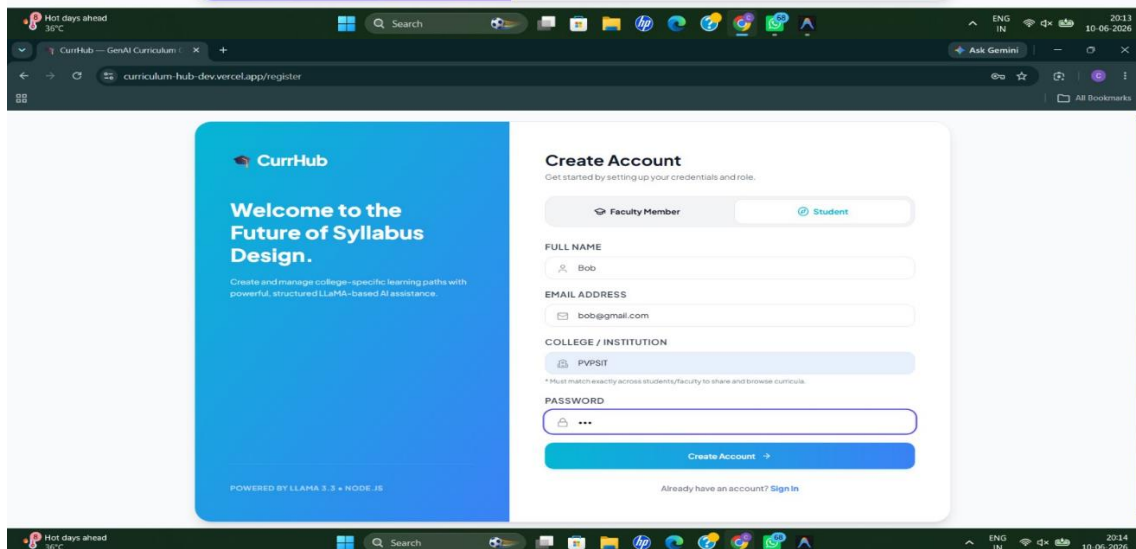
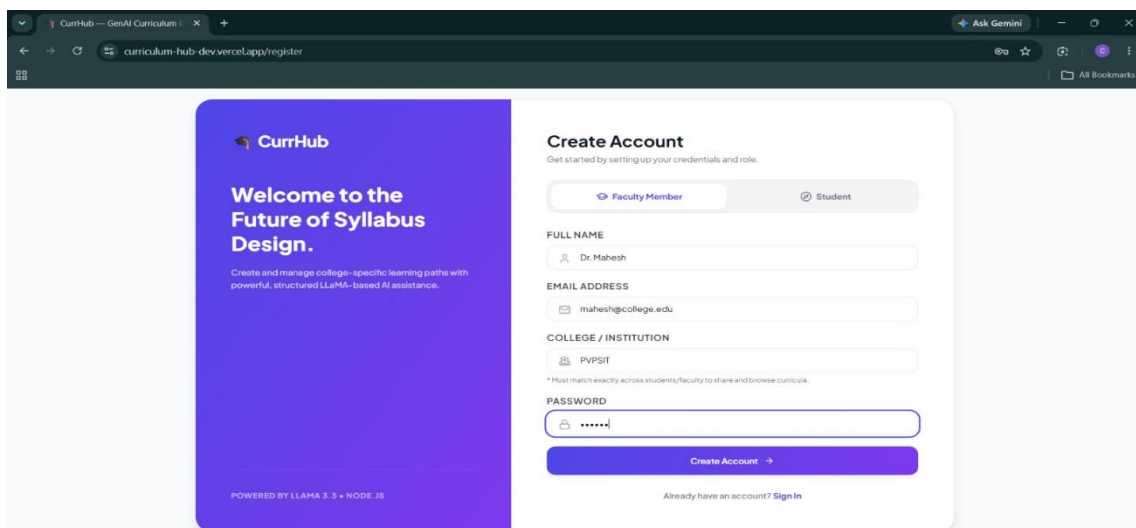
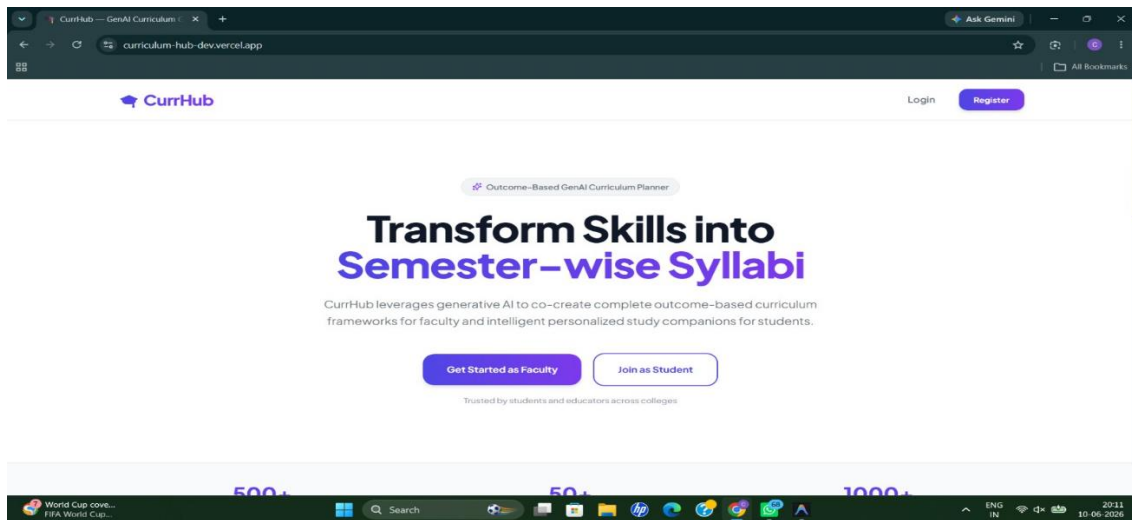
  if (!response.ok) {
    const errData = await response.json().catch(() => ({}));
    throw new Error(errData?.error?.message || `Groq API returned status ${response.status}`);
  }

  const data = await response.json();
  const resultText = data.choices[0].message.content;

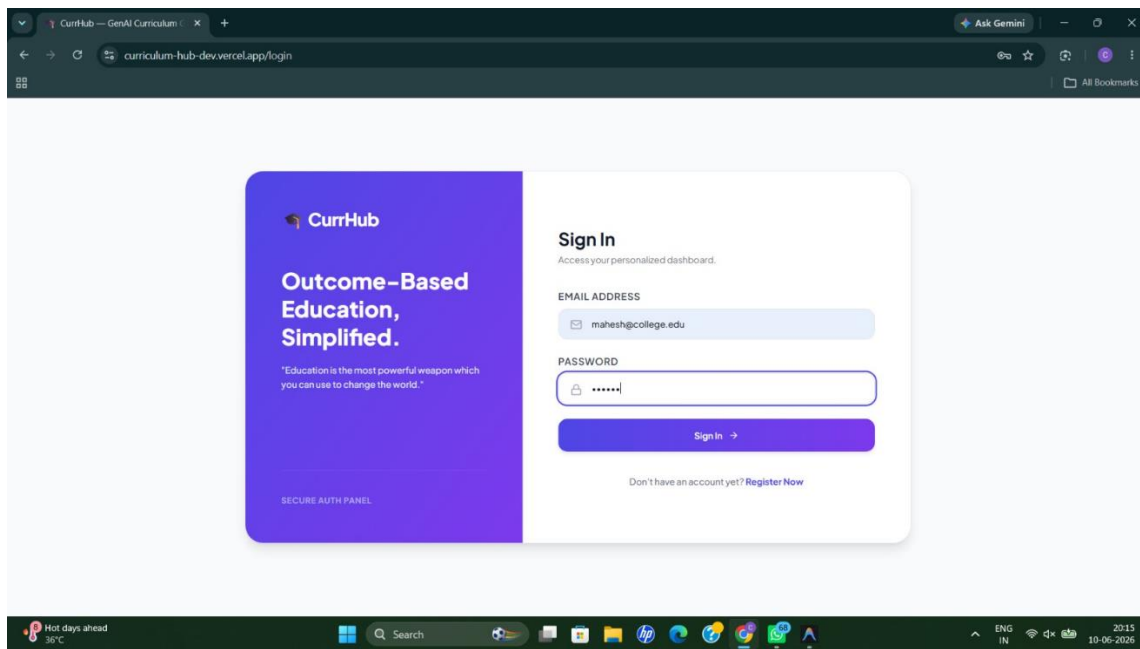
  // Parse response
  const parsedData = JSON.parse(resultText);
  return parsedData;
} catch (error) {
  console.error("Groq Curriculum Generation Error:", error);
  throw error;
}
}
```

Functional Testing:

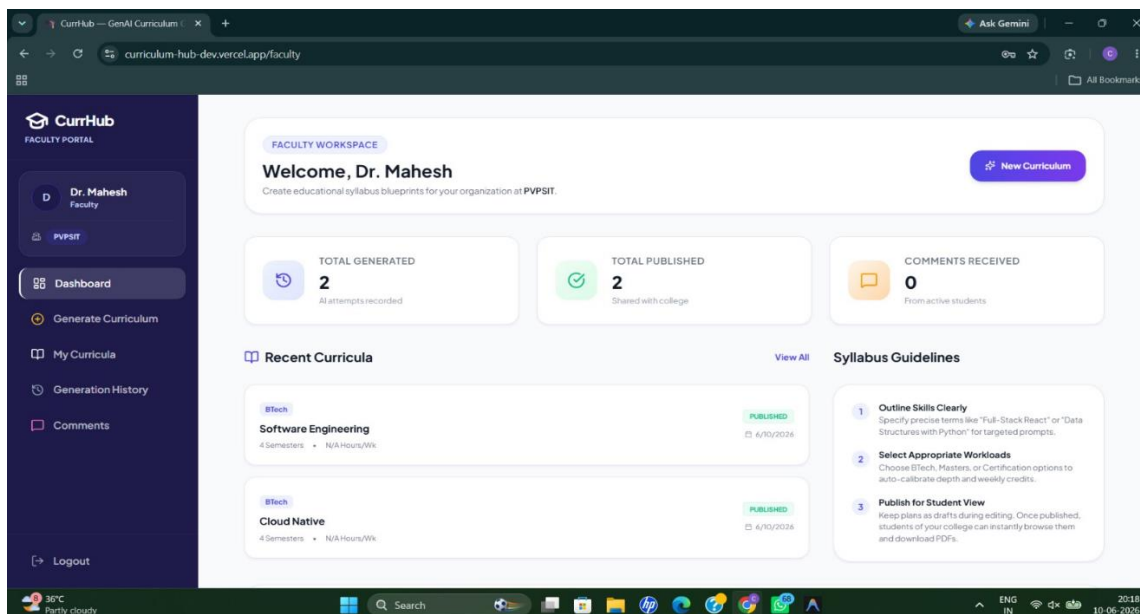
The application is tested to ensure all features operate correctly under different scenarios.



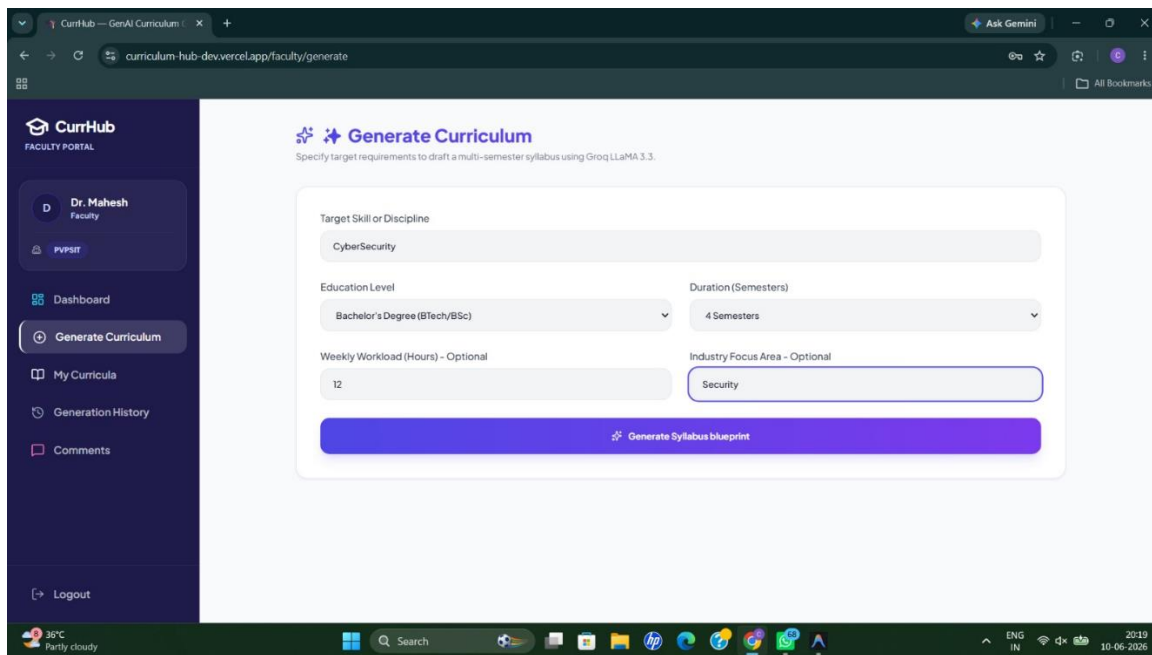
Sign in for student side:



The faculty member was successfully authenticated and granted access to the Faculty Dashboard with all curriculum management features enabled



Faculty Dashboard The main landing page for Dr. Mahesh at PVPSIT college, showing stats (2 generated, 2 published, 0 comments) and recently created curricula for Software Engineering and Cloud Native, along with syllabus guidelines on the right.



Generate Curriculum
Specify target requirements to draft a multi-semester syllabus using Groq LLaMA 3.3.

Target Skill or Discipline: CyberSecurity

Education Level: Bachelor's Degree (BTech/BSc)

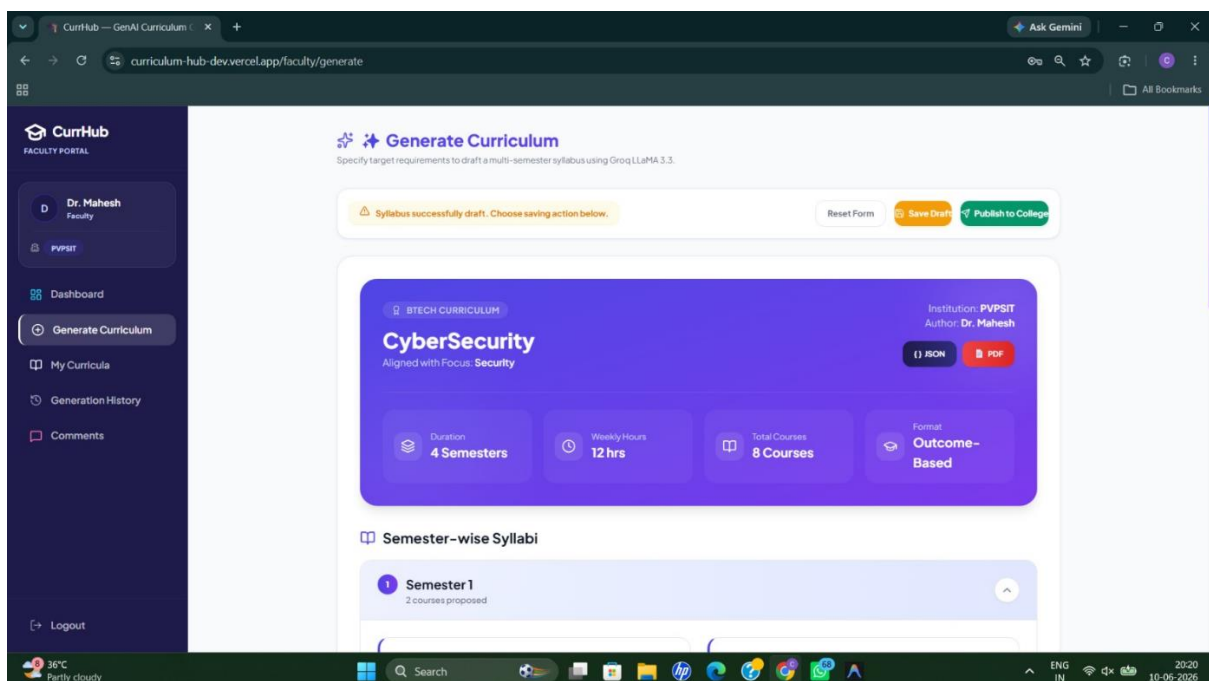
Duration (Semesters): 4 Semesters

Weekly Workload (Hours) - Optional: 12

Industry Focus Area - Optional: Security

Generate Syllabus blueprint

Generate Curriculum Form The curriculum creation form where faculty inputs details like Target Skill (CyberSecurity), Education Level (BTech), Duration (4 Semesters), Weekly Workload (12 hrs), and Industry Focus (Security) before clicking "Generate Syllabus Blueprint."



Generate Curriculum
Specify target requirements to draft a multi-semester syllabus using Groq LLaMA 3.3.

Syllabus successfully draft. Choose saving action below. [Reset Form](#) [Save Draft](#) [Publish to College](#)

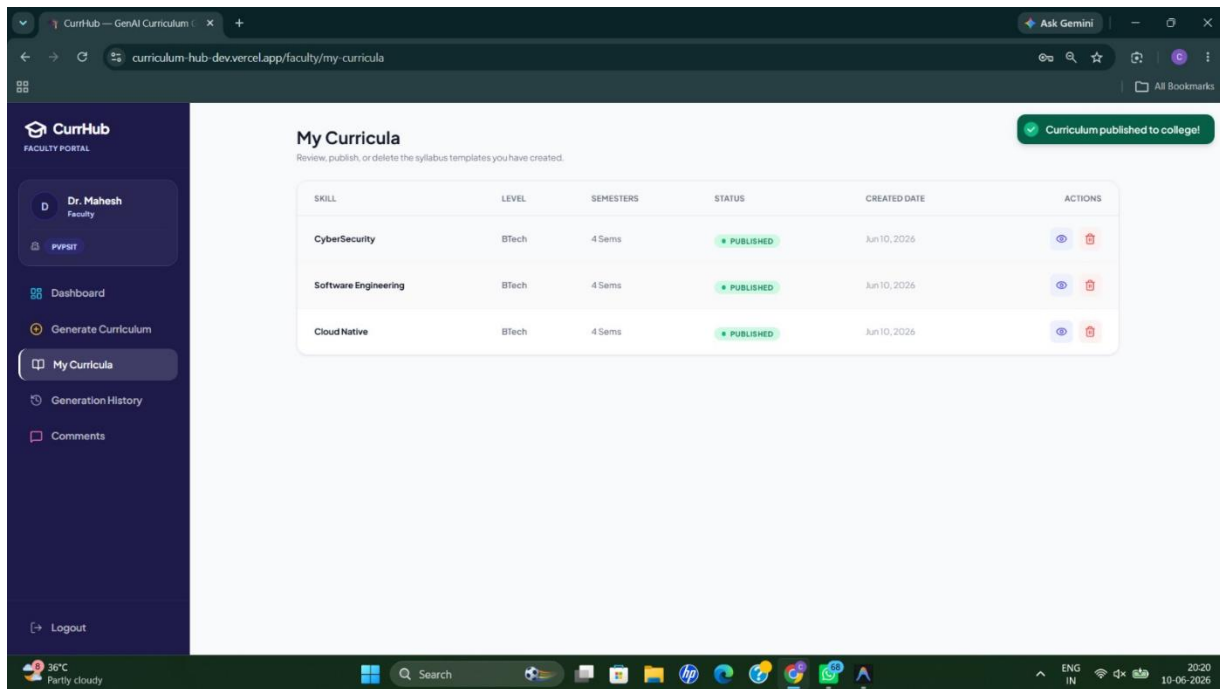
CyberSecurity
Institution: PVPISIT
Author: Dr. Mahesh
Aligned with Focus: Security

Duration: 4 Semesters | Weekly Hours: 12 hrs | Total Courses: 8 Courses | Format: Outcome-Based

Semester-wise Syllabi

Semester 1
2 courses proposed

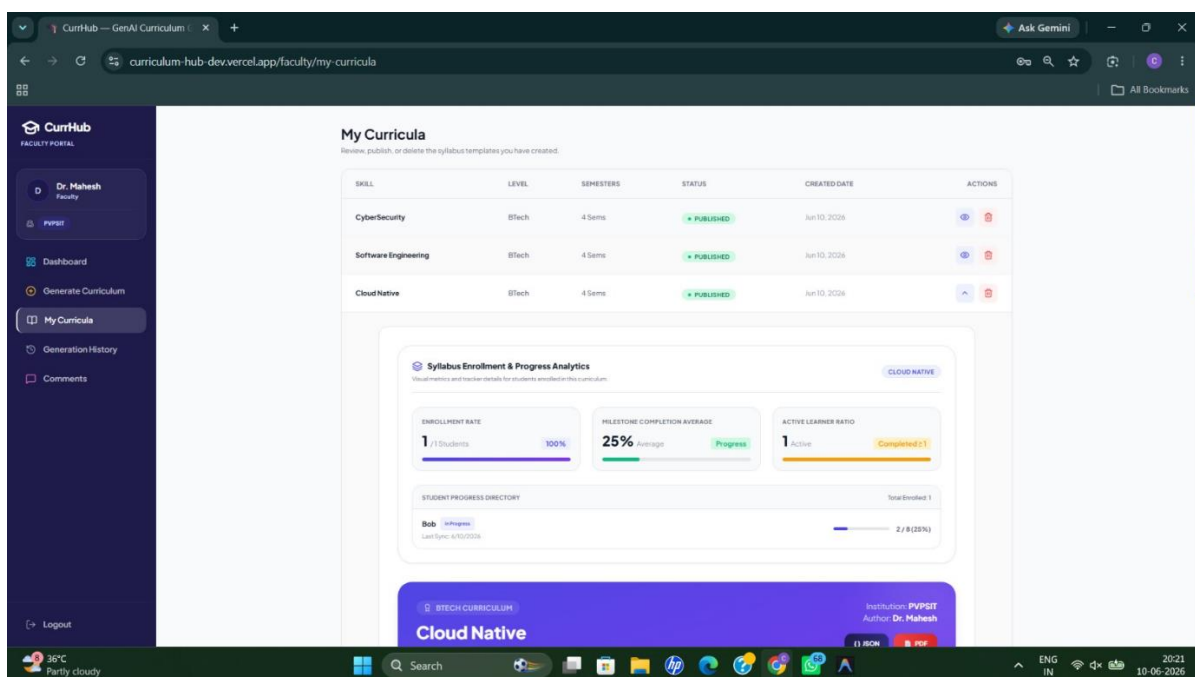
Generated Curriculum Preview After generation, the AI-drafted CyberSecurity BTech curriculum is displayed with a purple header card showing 4 Semesters, 12 hrs/week, 8 Courses in Outcome-Based format, with options to Save Draft, Publish to College, or export as JSON/PDF.



My Curricula
Review, publish, or delete the syllabus templates you have created.

SKILL	LEVEL	SEMESTERS	STATUS	CREATED DATE	ACTIONS
CyberSecurity	BTech	4 Sems	PUBLISHED	Jun 10, 2026	 
Software Engineering	BTech	4 Sems	PUBLISHED	Jun 10, 2026	 
Cloud Native	BTech	4 Sems	PUBLISHED	Jun 10, 2026	 

My Curricula List A table listing all three published curricula (CyberSecurity, Software Engineering, Cloud Native), all BTech level with 4 semesters, along with view and delete action buttons, confirming the CyberSecurity curriculum was just published.



My Curricula
Review, publish, or delete the syllabus templates you have created.

SKILL	LEVEL	SEMESTERS	STATUS	CREATED DATE	ACTIONS
CyberSecurity	BTech	4 Sems	PUBLISHED	Jun 10, 2026	 
Software Engineering	BTech	4 Sems	PUBLISHED	Jun 10, 2026	 
Cloud Native	BTech	4 Sems	PUBLISHED	Jun 10, 2026	 

Syllabus Enrollment & Progress Analytics
Visual metrics and real-time details for students enrolled in this curriculum.

ENROLLMENT RATE
1 / 1 Students 100%

MILESTONE COMPLETION AVERAGE
25% Average Progress

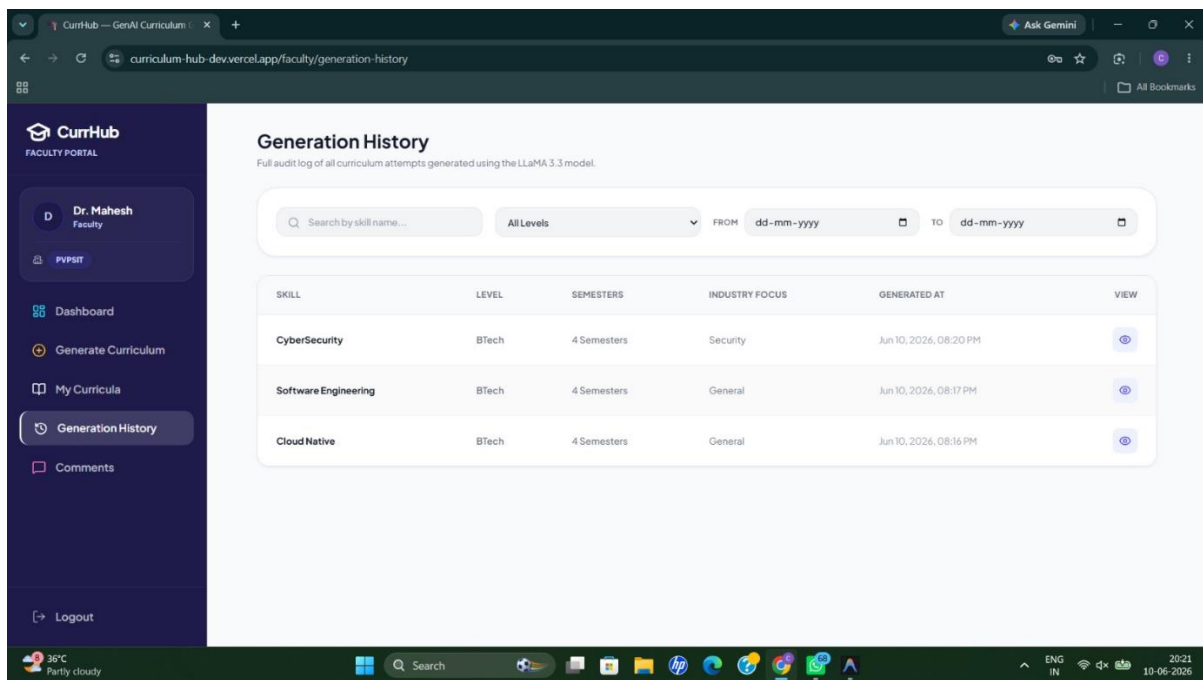
ACTIVE LEARNER RATIO
1 Active Completed 0

STUDENT PROGRESS DIRECTORY
Total Enrolled: 1

Bob In Progress
Last Sync: 6/10/2026 2 / 8 (25%)

Cloud Native
Institution: PVPIT
Author: Dr. Mahesh

Curriculum Analytics Panel Expanded view of the Cloud Native curriculum row showing Syllabus Enrollment & Progress Analytics — 1 student enrolled (Bob, In Progress at 25% completion), with enrollment rate, milestone completion, and active learner ratio metrics.

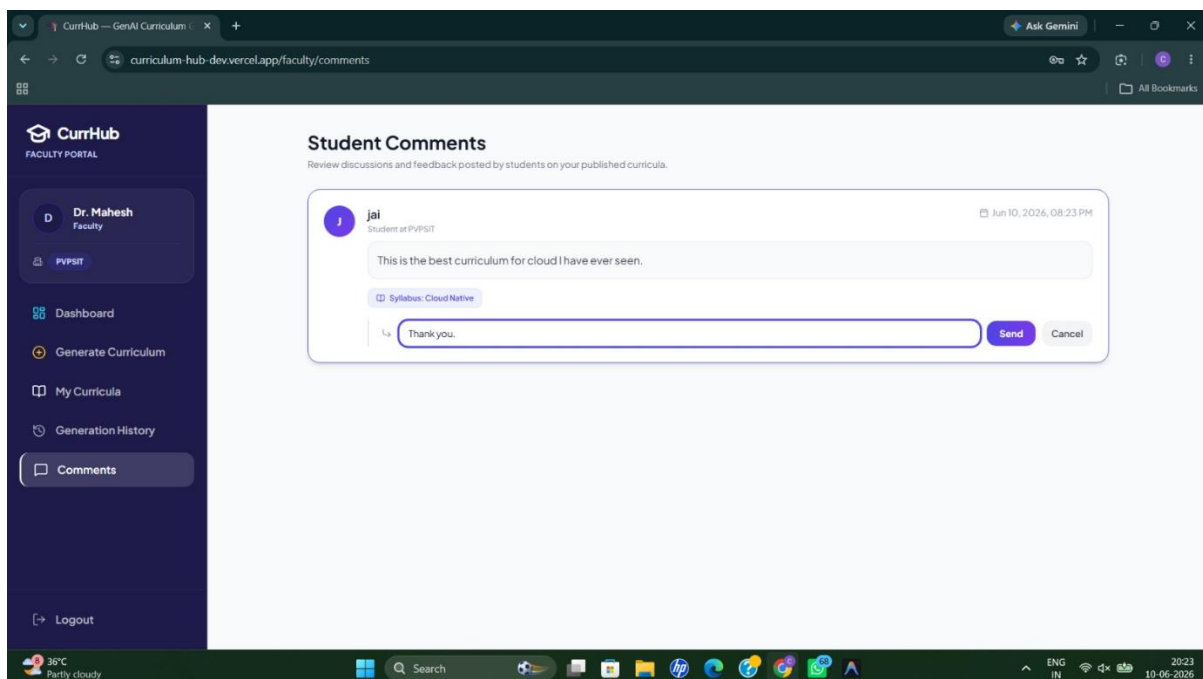


Generation History
Full audit log of all curriculum attempts generated using the LLaMA 3.3 model.

Search by skill name... All Levels FROM dd-mm-yyyy TO dd-mm-yyyy

SKILL	LEVEL	SEMESTERS	INDUSTRY FOCUS	GENERATED AT	VIEW
CyberSecurity	BTech	4 Semesters	Security	Jun 10, 2026, 08:20 PM	View
Software Engineering	BTech	4 Semesters	General	Jun 10, 2026, 08:17 PM	View
Cloud Native	BTech	4 Semesters	General	Jun 10, 2026, 08:16 PM	View

Generation History A full audit log of all AI generation attempts, listing CyberSecurity (Security focus), Software Engineering (General), and Cloud Native (General), each with BTech level, 4 semesters, and exact timestamps for faculty tracking.



Student Comments
Review discussions and feedback posted by students on your published curricula.

jai
Student at PVPSIT
Jun 10, 2026, 08:23 PM

This is the best curriculum for cloud I have ever seen.

Syllabus: Cloud Native

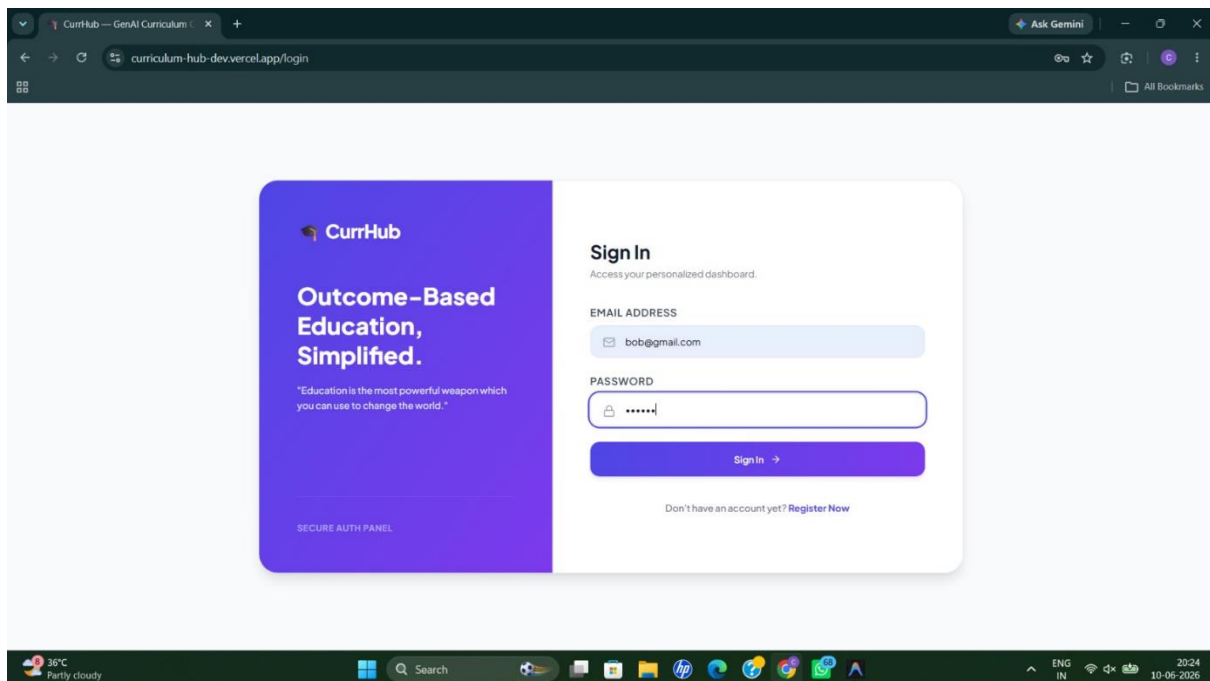
Thank you.

[Send](#) [Cancel](#)

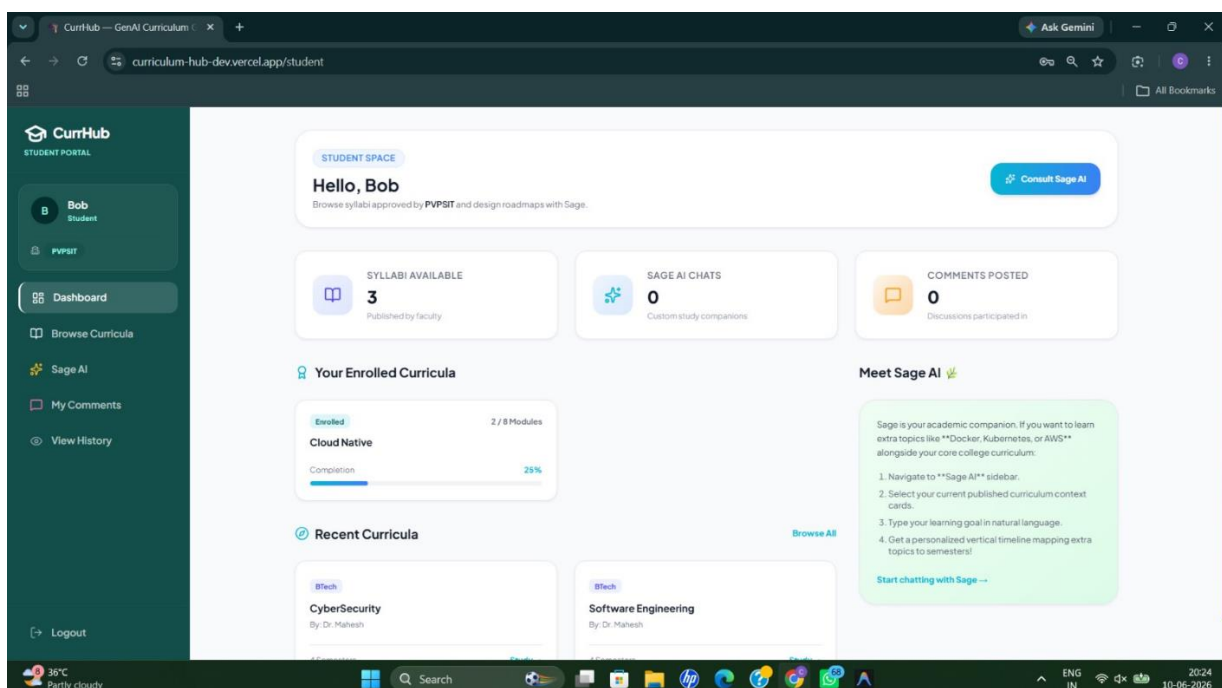
Student Comments The faculty comments page showing a student named "Jai" from PVPSIT praising the Cloud Native curriculum, with Dr. Mahesh typing a "Thank you." reply using the inline reply box linked to that specific syllabus.

Sign in for student side:

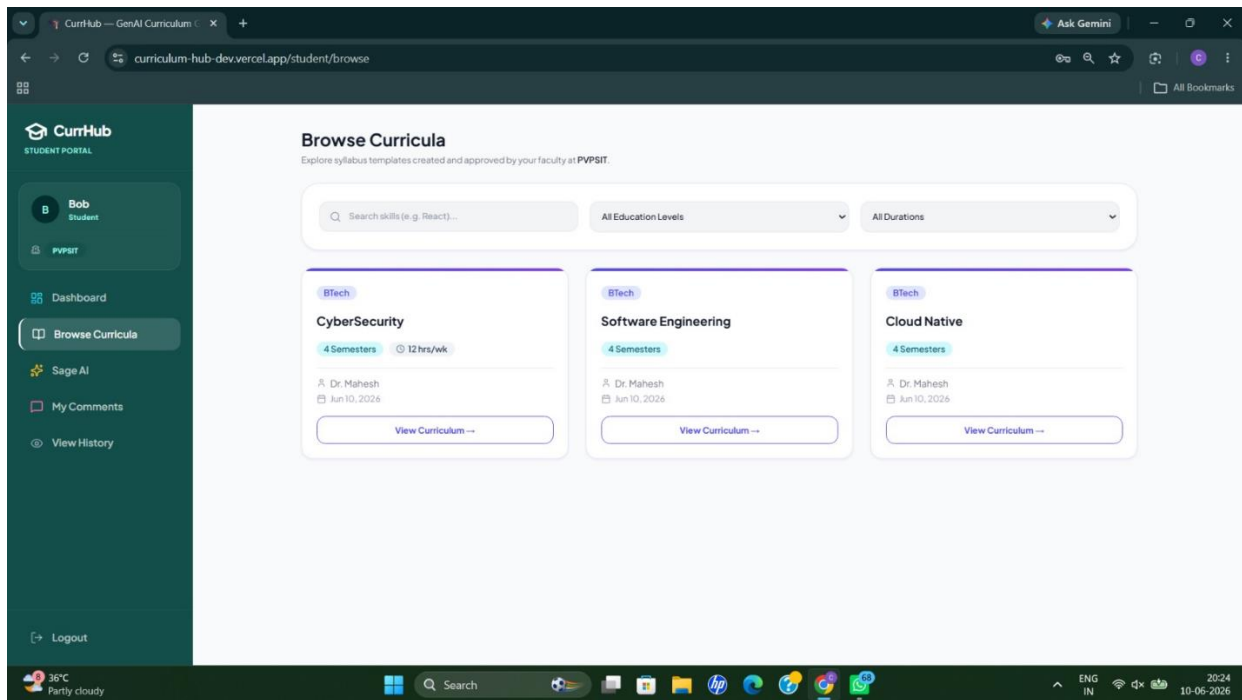
The student enters a valid email address and password to log in to the CurrHub platform and access personalized academic resources.



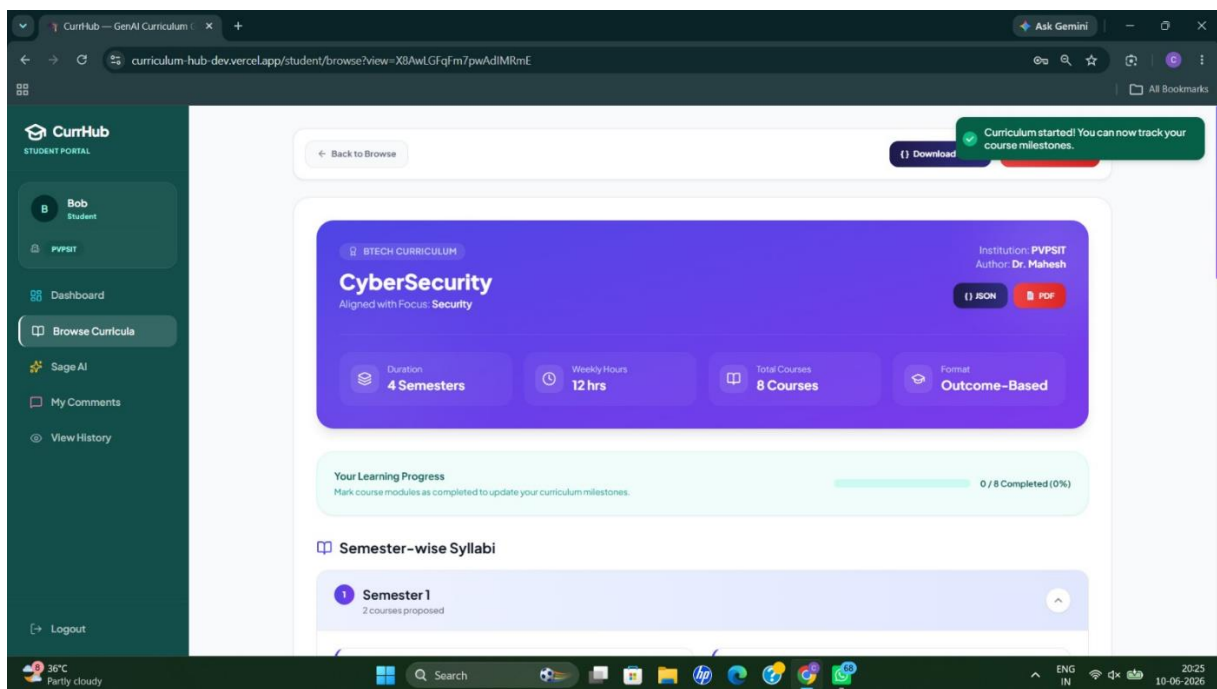
The student was successfully authenticated and redirected to the Student Dashboard. All student-specific features, including curriculum browsing and AI-assisted learning support, were accessible without any issues.



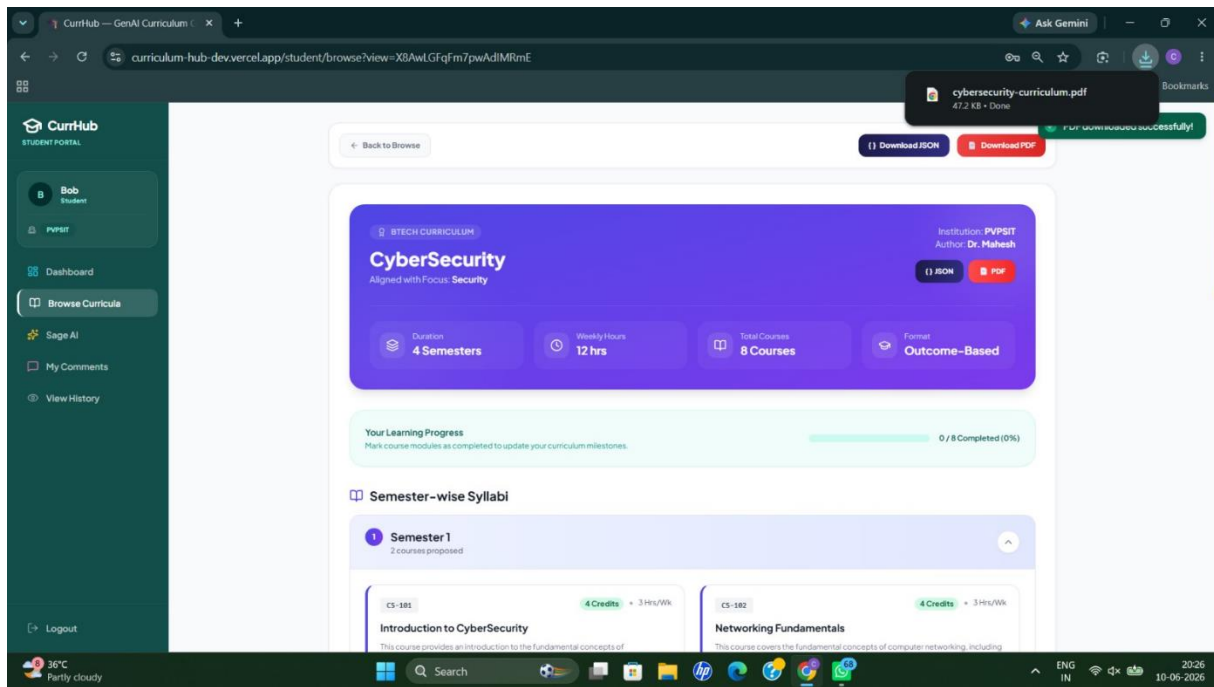
Student Dashboard The student landing page for Bob at PVPSIT, showing 3 syllabi available, 0 Sage AI chats, and 0 comments posted. It displays his enrolled Cloud Native curriculum at 25% completion (2/8 modules) and introduces the Sage AI companion feature on the right side.



Browse Curricula Page A searchable catalog of all faculty-published curricula at PVPSIT, displaying three B.Tech cards — CyberSecurity (12 hrs/wk), Software Engineering, and Cloud Native — each with 4 semesters, created by Dr. Mahesh, with a "View Curriculum" button on each card.



Curriculum Detail & Enrollment Bob has just enrolled in the CyberSecurity curriculum, confirmed by the green "Curriculum started!" toast notification. The purple header card shows 4 Semesters, 12 hrs/week, 8 Courses, Outcome-Based format, with a learning progress tracker showing 0/8 completed (0%) and semester-wise syllabi below.



currhub-dev.vercel.app/student/browseView-X8AwLGFqfm7pwAdIMRmE

cybersecurity-curriculum.pdf
47.2 KB • Done

PDF downloaded successfully!

CurrHub
STUDENT PORTAL

Bob Student
PVPSIT

Dashboard
Browse Curricula
Sage AI
My Comments
View History
Logout

← Back to Browse

BTECH CURRICULUM

CyberSecurity
Aligned with Focus: Security

Institution: PVPSIT
Author: Dr. Mahesh

Download JSON Download PDF

Duration: 4 Semesters
Weekly hours: 12 hrs
Total Courses: 8 Courses
Format: Outcome-Based

Your Learning Progress
Mark course modules as completed to update your curriculum milestones.
0 / 8 Completed (0%)

Semester-wise Syllabi

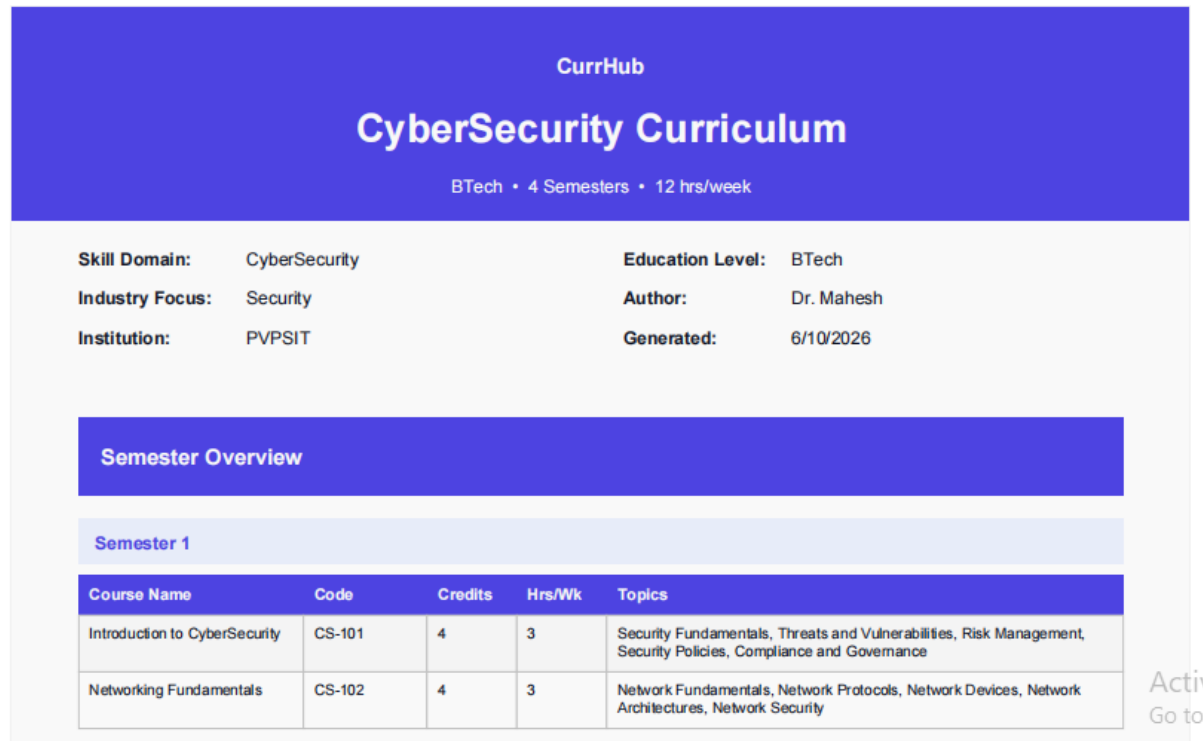
Semester 1
2 courses proposed

CS-101 4 Credits • 3 Hrs/Wk
Introduction to CyberSecurity
This course provides an introduction to the fundamental concepts of

CS-102 4 Credits • 3 Hrs/Wk
Networking Fundamentals
This course covers the fundamental concepts of computer networking, including

36°C Partly cloudy
Search
ENG IN
2026 10-06-2026

PDF Download of Curriculum Bob downloaded the CyberSecurity curriculum as a PDF (cybersecurity-curriculum.pdf, 47.2 KB), confirmed by the "PDF downloaded successfully!" toast. The Semester 1 syllabi shows two courses: CS-101 Introduction to CyberSecurity and CS-102 Networking Fundamentals, each with 4 Credits and 3 hrs/week.



CurrHub

CyberSecurity Curriculum

BTech • 4 Semesters • 12 hrs/week

Skill Domain: CyberSecurity
Industry Focus: Security
Institution: PVPSIT

Education Level: BTech
Author: Dr. Mahesh
Generated: 6/10/2026

Semester Overview

Semester 1

Course Name	Code	Credits	Hrs/Wk	Topics
Introduction to CyberSecurity	CS-101	4	3	Security Fundamentals, Threats and Vulnerabilities, Risk Management, Security Policies, Compliance and Governance
Networking Fundamentals	CS-102	4	3	Network Fundamentals, Network Protocols, Network Devices, Network Architectures, Network Security

ActiveGo to

Semester 2

Course Name	Code	Credits	Hrs/Wk	Topics
Cryptography and Access Control	CS-201	4	3	Cryptography Fundamentals, Encryption and Decryption, Access Control, Authentication and Authorization, Digital Signatures
Operating System Security	CS-202	4	3	Operating System Security Fundamentals, Secure Boot and Firmware, Secure Software Development, Operating System Hardening, Vulnerability Management

Semester 3

Course Name	Code	Credits	Hrs/Wk	Topics
Network Security	CS-301	4	3	Network Security Fundamentals, Firewalls and Network Segmentation, Intrusion Detection and Prevention Systems, Virtual Private Networks, Network Security Protocols
Web Application Security	CS-302	4	3	Web Application Security Fundamentals, Secure Coding Practices, Input Validation and Sanitization, Authentication and Authorization, Web Application Security Protocols

Semester 4

Course Name	Code	Credits	Hrs/Wk	Topics
Incident Response and Disaster Recovery	CS-401	4	3	Incident Response Fundamentals, Incident Response Planning, Disaster Recovery Planning, Business Continuity Planning, Incident Response and Disaster Recovery Protocols
Cybersecurity Governance and Compliance	CS-402	4	3	Cybersecurity Governance Fundamentals, Cybersecurity Frameworks, Regulations and Standards, Compliance and Risk Management, Cybersecurity Governance and Compliance Protocols

Course Details

In-depth descriptions and learning outcomes for every course

Introduction to CyberSecurity

CS-101

Semester 1 • 4 Credits • 3 Hrs/Week

DESCRIPTION

This course provides an introduction to the fundamental concepts of cybersecurity, including security threats, vulnerabilities, and risk management.

LEARNING OUTCOMES

- Understand the basic concepts of cybersecurity
- Identify security threats and vulnerabilities
- Apply risk management principles
- Develop security policies and procedures

Networking Fundamentals

CS-102

Semester 1 • 4 Credits • 3 Hrs/Week

DESCRIPTION

This course covers the fundamental concepts of computer networking, including network protocols, devices, and architectures.

LEARNING OUTCOMES

- Understand network fundamentals
- Identify network protocols and devices
- Apply network architectures and security principles
- Configure network devices and protocols

Cryptography and Access Control

CS-201

Semester 2 • 4 Credits • 3 Hrs/Week

DESCRIPTION

This course covers the fundamental concepts of cryptography and access control, including encryption, decryption, and authentication.

LEARNING OUTCOMES

- Understand cryptography fundamentals
- Apply encryption and decryption techniques
- Implement access control and authentication mechanisms
- Develop secure authentication and authorization protocols

Operating System Security

CS-202

Semester 2 • 4 Credits • 3 Hrs/Week

DESCRIPTION

This course covers the fundamental concepts of operating system security, including secure boot, secure firmware, and secure software development.

LEARNING OUTCOMES

- Understand operating system security fundamentals
- Apply secure boot and firmware principles
- Develop secure software development practices
- Implement operating system hardening and vulnerability management

Network Security

CS-301

Semester 3 • 4 Credits • 3 Hrs/Week

DESCRIPTION

This course covers the fundamental concepts of network security, including firewalls, intrusion detection systems, and virtual private networks.

LEARNING OUTCOMES

- Understand network security fundamentals
- Apply firewall and network segmentation principles
- Implement intrusion detection and prevention systems

Incident Response and Disaster Recovery

CS-401

Semester 4 • 4 Credits • 3 Hrs/Week

DESCRIPTION

This course covers the fundamental concepts of incident response and disaster recovery, including incident response planning, disaster recovery planning, and business continuity planning.

LEARNING OUTCOMES

- Understand incident response fundamentals
- Apply incident response planning principles
- Implement disaster recovery planning and business continuity planning
- Develop incident response and disaster recovery protocols

Cybersecurity Governance and Compliance

CS-402

Semester 4 • 4 Credits • 3 Hrs/Week

DESCRIPTION

This course covers the fundamental concepts of cybersecurity governance and compliance, including cybersecurity frameworks, regulations, and standards.

LEARNING OUTCOMES

- Understand cybersecurity governance fundamentals
- Apply cybersecurity frameworks and regulations
- Implement compliance and risk management practices
- Develop cybersecurity governance and compliance protocols

Capstone Project

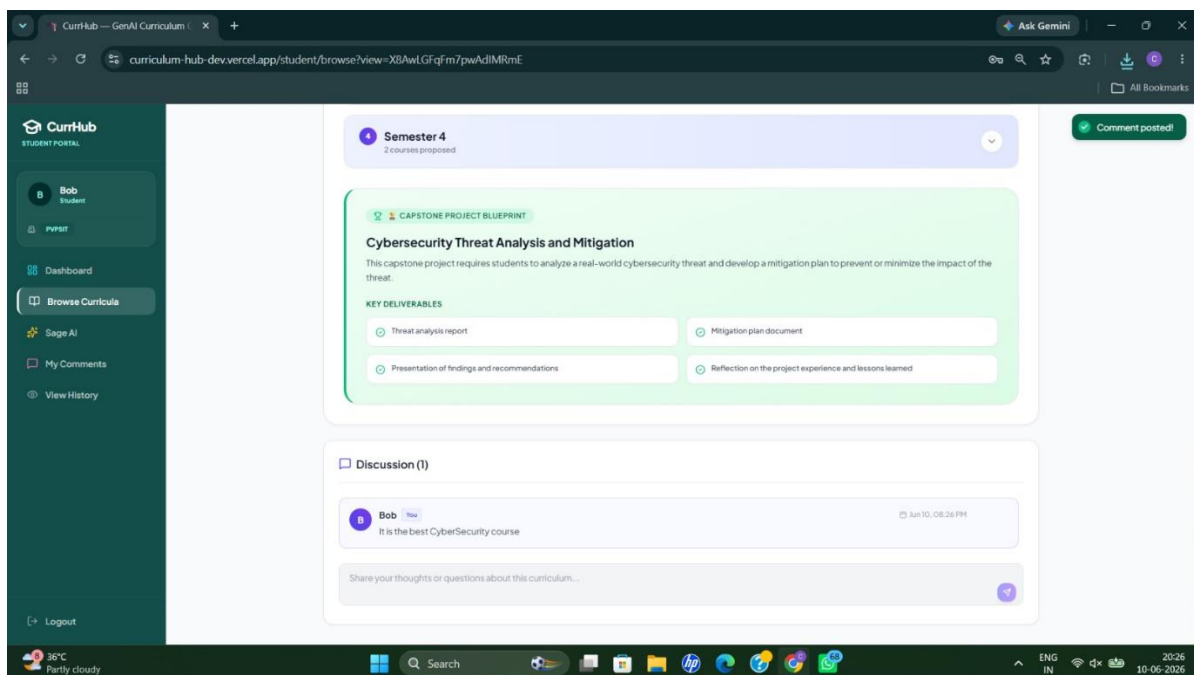
Final integrative project synthesizing the full curriculum

Cybersecurity Threat Analysis and Mitigation

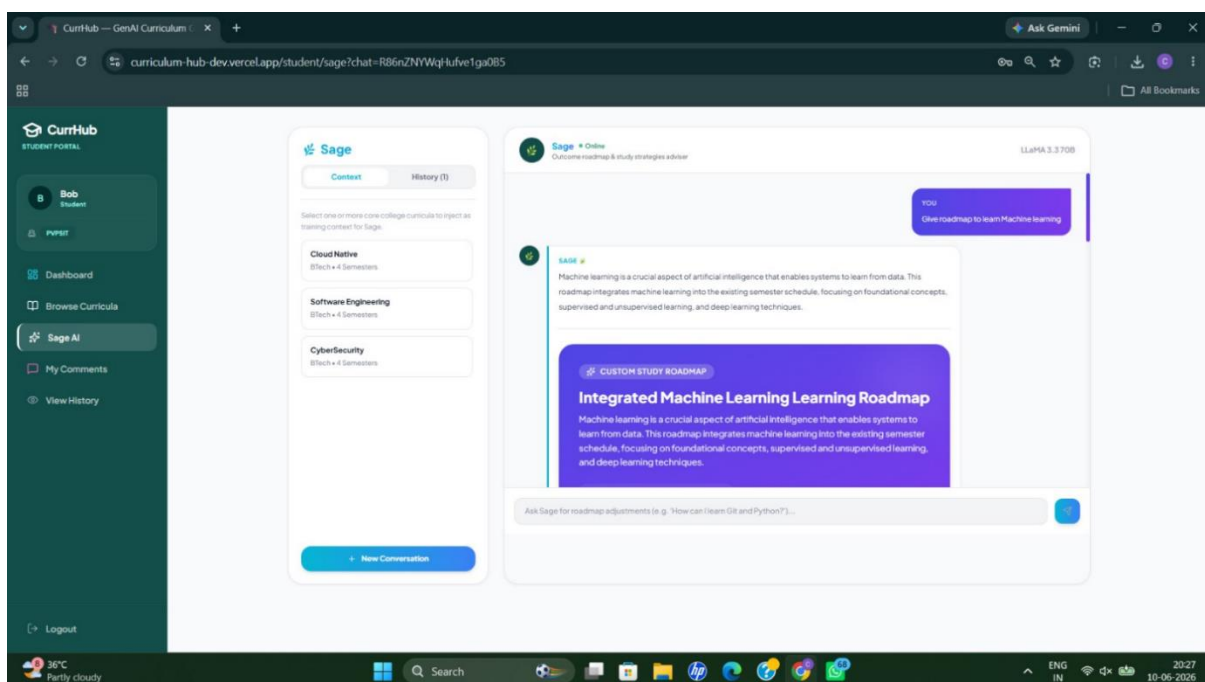
This capstone project requires students to analyze a real-world cybersecurity threat and develop a mitigation plan to prevent or minimize the impact of the threat.

KEY DELIVERABLES

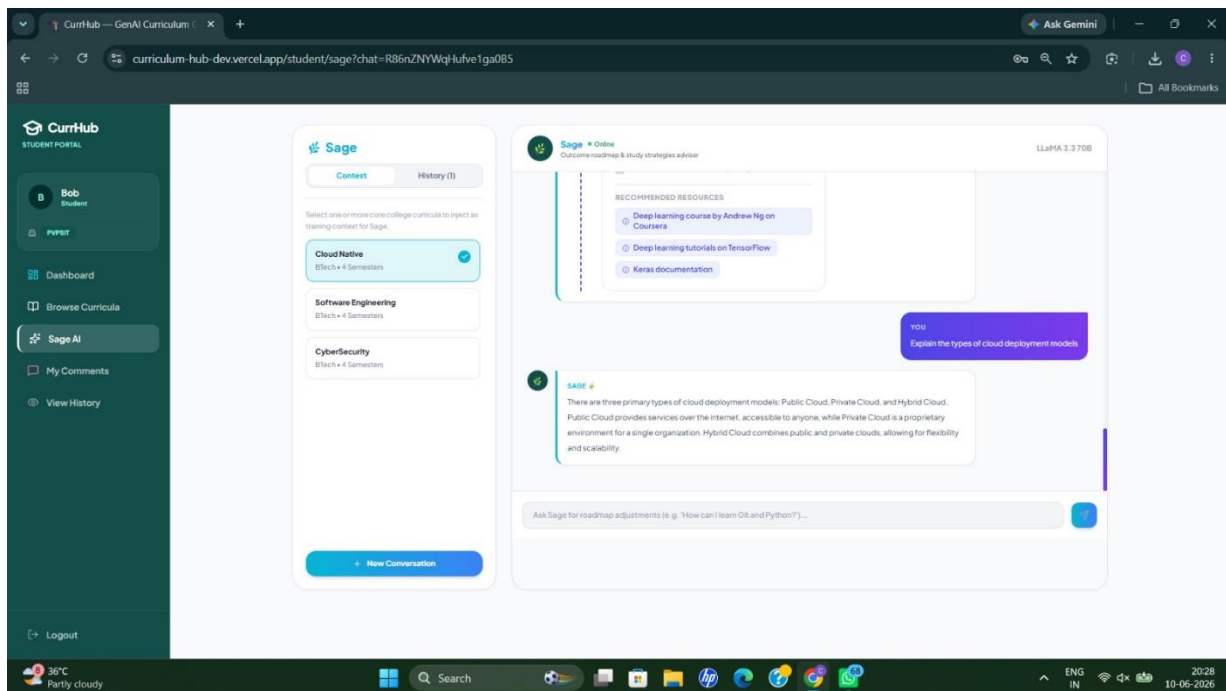
- 1 Threat analysis report
- 2 Mitigation plan document
- 3 Presentation of findings and recommendations
- 4 Reflection on the project experience and lessons learned



apstone Project & Discussion Section The bottom of the CyberSecurity curriculum page showing the Semester 4 Capstone Project Blueprint titled "Cybersecurity Threat Analysis and Mitigation," with four key deliverables. Below it, the Discussion section shows Bob's posted comment "It is the best CyberSecurity course," confirmed by the green "Comment posted!" toast.



Sage AI – Custom Study Roadmap Bob asked Sage AI (powered by LLaMA 3.37B) to "Give roadmap to learn Machine learning." Sage responded with a custom study roadmap card titled "Integrated Machine Learning Learning Roadmap," integrating ML topics like foundational concepts, supervised/unsupervised learning, and deep learning into the existing semester schedule.



Sage AI – Cloud Deployment Explanation Continuing the Sage AI chat with Cloud Native curriculum selected as context, Bob asked Sage to "Explain the types of cloud deployment models." Sage responded with a clear explanation of the three types — Public Cloud, Private Cloud, and Hybrid Cloud — along with recommended resources like Andrew Ng's Deep Learning course on Coursera, TensorFlow tutorials, and Keras documentation.

MILESTONE 5: Deployment & Documentation

Activity 5.1: Application Deployment

Description

The CurrHub application was successfully deployed using the Vercel cloud hosting platform. Vercel provides automated build, deployment, and hosting services for modern web applications. The deployment process ensures that the application is accessible online, scalable, and available to users without requiring local installation.

Deployment Process

1. The source code was developed using React.js and Vite.
2. The project repository was pushed to GitHub.
3. The GitHub repository was connected to the Vercel platform.
4. Vercel automatically detected the Vite configuration and executed the build process.
5. Environment variables for Firebase and Groq API were configured in Vercel.
6. After successful compilation, the application was deployed and made publicly accessible.

Live Application URL

The deployed CurrHub application can be accessed using the following URL:

<https://curriculum-hub-dev.vercel.app>

Deployment Outcome

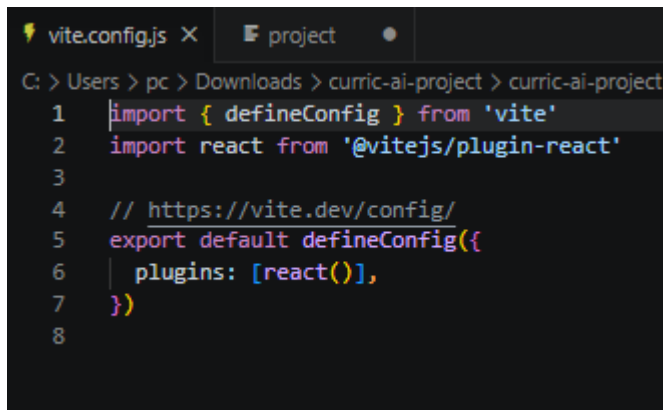
The deployed application enables faculty members to generate AI-powered curricula, manage curriculum data, and publish academic content. Students can browse published curricula, interact with the Sage AI assistant, download curriculum PDFs, and access personalized learning resources through a web browser from any location.

Files Used

- **package.json**
- **vite.config.js**

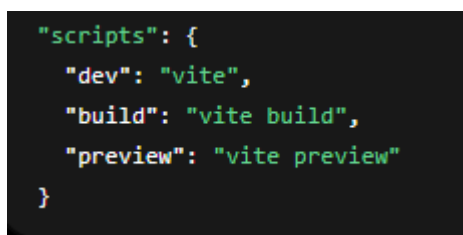
Deployment Configuration (vite.config.js)

Configures the Vite build system and React plugin required for deployment.



Build Script (package.json)

Defines commands for development, production build generation, and application preview.



Activity 5.2: Documentation

Comprehensive documentation was prepared to assist users and developers in understanding, installing, configuring, and operating the CurrHub platform. The documentation provides information about project features, technology stack, installation procedures, and usage guidelines.

File Used

README.md

Installation Guide

1. Clone the CurrHub repository from GitHub.
2. Navigate to the project directory.
3. Install all required dependencies using:

npm install

4. Configure Firebase and Groq API credentials in the .env file.
5. Start the development server using:

npm run dev

6. Open the application in a web browser.

Features

- AI-powered curriculum generation using LLaMA 3.3 70B.
- Faculty curriculum creation and management.
- Student curriculum browsing and curriculum viewing.
- Sage AI learning roadmap assistant.
- Curriculum PDF export functionality.
- Real-time comments and feedback system.
- Curriculum generation history tracking.
- Secure authentication and role-based access control.

Technology Stack

Frontend Technologies:

- React.js
- Tailwind CSS
- React Router

Backend Services:

- Firebase Authentication
- Firebase Firestore Database

Artificial Intelligence:

- Groq API
- LLaMA 3.3 70B Model

Deployment Platform:

- Vercel

Additional Libraries:

- jsPDF
- jspdf-autotable
- Lucide React

Usage Instructions

Faculty Users:

1. Register and log in as Faculty.
2. Generate AI-powered curricula.
3. Save and publish generated curricula.
4. Manage curriculum history.
5. Monitor student comments and feedback.

Student Users:

1. Register and log in as Student.
2. Browse published curricula.
3. View detailed curriculum information.
4. Download curriculum PDFs.
5. Interact with Sage AI for personalized learning roadmaps.
6. View learning history and comments.

Project Repository

Live Application

Vercel Deployment URL:

<https://curriculum-hub-dev.vercel.app>

Conclusion

CurrHub is an AI-powered curriculum generation and management platform designed to simplify and modernize the educational planning process. By leveraging Generative AI through the Groq API, the system enables faculty members to create structured, industry-relevant curricula efficiently while providing students with easy access to learning resources, curriculum feedback, and personalized study roadmaps through Sage AI.

The platform successfully integrates modern technologies such as React, Firebase, Tailwind CSS, and Groq AI to deliver a seamless, scalable, and user-friendly experience. Features such as curriculum generation, publication management, real-time comments, PDF export, and AI-guided learning support make CurrHub a comprehensive solution for academic institutions.

By reducing the time and effort required for curriculum design and improving collaboration between faculty and students, CurrHub demonstrates the potential of Artificial Intelligence in transforming education. The project serves as a strong foundation for future enhancements and smart educational ecosystems.

Live Application: <https://curriculum-hub-dev.vercel.app>

In conclusion, CurrHub bridges the gap between academic curriculum development and industry requirements, providing an intelligent, efficient, and collaborative platform that enhances both teaching and learning experiences.